

Master Diploma Orthopaedics Admission Preparation Guide

Bangladesh 2027 Edition — Top 100 Sureshot Topics

BMU / BSMMU Diploma & MPhil (July 2026) · Residency (March 2026)

A pure suggestion & rapid-revision book for the Diploma in Orthopaedics (D.Ortho) entrance MCQ test

Top 100 sureshot topics · Basic Sciences · Faculty Surgery · Orthopaedics High-Yield · SBA/MCQ · Rapid Revision · Exam-Night Chart · 3-Month Plan

Standard references: Apley · Campbell · Rockwood & Green · Miller's Review · AO Principles · Bailey & Love · Schwartz · Gray's · Snell · Ganong · Guyton · Robbins · Katzung · Lippincott

Compiled: July 2027 · Probability-rated (very-high → rare) · For self-study use only — verify against current BMU/BSMMU notice & named textbooks before the exam.

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HOW TO USE THIS BOOK · STRUCTURE · PROBABILITY SYSTEM

Master Diploma Orthopaedics Admission Preparation Guide — Bangladesh 2027 Edition

Who this is for

MBBS graduates sitting the **Diploma in Orthopaedics (D.Ortho)** entrance MCQ test (BMU/BSMMU, July 2026 / 2027 session) and benchmarking against BSMMU Residency (Orthopaedics) MCQ. It is a *suggestion + rapid-revision* book, **not** a substitute for reading the standard textbooks listed at the end.

Book structure (7 parts, built in stages)

1. **Basic Sciences** — Anatomy, Physiology, Biochemistry, Pathology, Pharmacology, Microbiology (25 topics)
2. **Faculty Surgery** — trauma, wounds, vascular, hand, amputation, gait (20 topics)
3. **Orthopaedics High-Yield** — fractures, pediatric, spine, joints, sports, tumors, infection (45 topics)
4. **Top Repeating Common Questions** — the "10000% common" cross-cutting facts (10 topics)
5. **Rapid Revision + Predicted High-Yield + Image-based + Flowcharts + Exam-Night Chart**
6. ■ **3-Month Study Plan** — for a sure chance in the admission test
7. **Print-Ready PDF + Copy-Paste Notes**

Parts 1–4 together form the **TOP 100 SURESHOT TOPICS** (25 + 20 + 45 + 10 = 100).

★ Probability (Yield) System

Every topic is ranked by likelihood of appearing, based on:

(a) previous-question repetition, (b) syllabus weighting, (c) recurrence across BMU/BSMMU/residency banks.

Icon	Level	Meaning	Revision priority
★★★★★	Very High Yield	Repeated almost every year / core concept	Must master first
★★★★	High Yield	Appears frequently	Must know
★★★	Moderate	Seen intermittently	Should know
★★	Low	Rare but possible	Glance before exam
★	Rare	Long-shot	Only if time permits

Per-topic output format (used in Parts 1–4)

Each topic block contains:

- **High-Yield Notes** (concise facts)
- **Rapid Revision** (one-line recalls)
- **Repeating SBA + MCQ** (2–3 stems)
- **Detailed Explanation** (why the answer is right)

- **Previous-Question Trend** (how it has appeared)
- **Clinical Correlation**
- **Examiner Tips** (trap answers)
- **Reference** (textbook + chapter)

Standard References

Orthopaedics: Apley's System of Orthopaedics & Fractures · Campbell's Operative Orthopaedics · Rockwood & Green's Fractures · Miller's Review of Orthopaedics · Netter's Atlas · AO Principles of Fracture Management

Surgery: Bailey & Love · Schwartz · Sabiston

Basic Sciences: Gray's Anatomy · Snell Clinical Anatomy · Ganong · Guyton & Hall · Robbins Basic Pathology · Katzung Pharmacology · Lippincott Microbiology

Bangladesh Sources: BMU/BSMMU Diploma & MPhil syllabus/notices · Residency admission syllabus · Previous admission question trends · NITOR academic resources

■ ■ Honest disclaimer

"10000% common" / "sure common" are the **candidate community's** framing. This book ranks topics by *evidence-based probability*, not guarantees. Use probability as a study-priority tool, not a promise. Verify any fact you are unsure of against the named reference before the exam.

Recommended reading loop

1. Skim `04-TOP-100-INDEX.md` → know the map.
2. Study ★★★★★ topics first (Parts 1–4).
3. Drill `08-SBA-MCQ-BANK.md` daily (20 Q/day).
4. Final 14 days → `09-RAPID-REVISION-EXAM-NIGHT.md` only.

SYLLABUS & EXAM-PATTERN ANALYSIS

BMU/BSMMU Diploma & MPhil (July 2026) · BSMMU Residency (March 2026)

1. The reality of the "syllabus"

BMU/BSMMU **does not publish a fixed topic-by-topic syllabus** for the Diploma or Residency entrance. The admission notice (e.g., *BMU Admission Notice for M.Phil, M.MEd & Diploma Courses — July 2026*, bmu.ac.bd/notices) specifies eligibility, seats, exam format and dates — **not** a content list. Preparation therefore mirrors the MBBS curriculum weighted toward Basic Sciences + Faculty Surgery + Orthopaedics, plus the documented question-bank trends.

2. Exam-format analysis (synthesised from notices + banks)

Parameter	BMU/BSMMU Diploma (D.Ortho)	BSMMU Residency (Ortho) MCQ
Type	MCQ / SBA	MCQ / SBA
Total marks	~100	~100–200
Duration	~1 hour	~2–3 hours
Negative marking	Usually none / minimal (verify per notice)	Verify per notice
Domain split	Basic Sciences + Surgery + Orthopaedics	Same, deeper ortho
Style	Single-best-answer, "EXCEPT", "Most common", "Best next step"	Same + image-based

Action: Always re-read the specific year's notice for marks, duration and negative-marking before you sit. The pattern above is the stable historical norm.

3. Implied subject weightage (from repeated-bank analysis)

Block	Est. weight	Notes
Basic Sciences (Anat/ Physio/Biochem/Path/ Pharma/Micro)	30–35%	Anatomy & Pathology dominate
Faculty Surgery (trauma, wounds, vascular, hand, amputation)	25–30%	Trauma/ortho-adjacent surgery heavy
Orthopaedics (fractures, pediatrics, spine, joints, tumors, infection)	35–40%	The "decider" block
Image-based / X-ray / photo	5–10%	Rising trend

4. BMU Diploma vs BSMMU Residency — key differences

- **Diploma:** Broader basic-science & general-surgery coverage; ortho depth moderate.
- **Residency:** Heavier ortho sub-specialty, more clinical-scenario SBAs, more image-based X-ray interpretation, higher cut-off.
- **Overlap (use it):** Fracture classifications (Garden, Gustilo, Salter-Harris, Weber/Lauge-Hansen), nerve injuries, compartment syndrome, osteomyelitis, AVN, pediatric hip, spine TB, OA/RA — appear in **both**.

5. Documented pattern features (Bangladesh banks)

- **Classification questions** are the single most repeated format (Garden I–IV, Gustilo I–III, Salter-Harris I–V, Weber A/B/C, Schatzker, Evans, Gartland, Cobb).
- **"Most common..."** stems dominate (organism, site, tumor, fracture, nerve injury).
- **EXCEPT / NOT true** stems test precision — high discriminator.
- **Image-based** (X-ray of Colle's, scaphoid, NOF, supracondylar, scoliosis) rising.
- **Numericals** occasionally tested (Bohler's angle >20°, Cobb >10°, Schober <4 cm).

6. What raises yield in the Bangladesh context (epidemiology)

- **Road-traffic trauma** → open fractures, femoral/pelvic, polytrauma.
- **Spinal TB (Pott's)** → cold abscess, paraplegia — highly tested in BD.
- **Vitamin-D deficiency / rickets / osteomalacia** → endemic.
- **Osteomyelitis (paediatric haematogenous)** → femur/metaphysis.
- **Diabetic foot / Charcot joint** → rising non-communicable burden.

7. Strategic implication

Weight your 3-month plan as **Ortho 40% · Basic Sciences 33% · Surgery 27%**, with a daily 20-question SBA drill and a weekly image/X-ray session. This matches the observed weightage and the BMU/BSMMU pattern.

PREVIOUS-QUESTION TREND ANALYSIS

Diploma / Residency Orthopaedics entrance — Bangladesh (synthesised from banks & examiner lore)

1. Recurrence law

In Bangladesh PG/Diploma ortho entrances, **~60–70% of the paper is recycled or variation-tested material** from prior years. The remaining 30–40% are new stems on classic high-yield concepts. Mastering repeats is the highest-ROI activity.

2. Most-repeated TOPIC CLUSTERS (by frequency)

Rank	Cluster	Typical stem style	Probability
1	Fracture classifications (Garden, Gustilo, Salter-Harris, Weber, Gartland, Schatzker, Evans, Cobb)	"Type II Gustilo is characterised by..."	★★★★★
2	Femoral neck / intertrochanteric / shaft femur	"Most common complication of NOF #" (AVN/nonunion)	★★★★★
3	Paediatric hip (DDH, Perthes, SCFE)	"Ortolani/BARLOW positive in..."	★★★★★
4	Osteomyelitis & septic arthritis	"Most common organism" (S. aureus)	★★★★★
5	Nerve injuries (radial, median, common peroneal, sciatic)	"Wrist drop is due to lesion of..."	★★★★★
6	Compartment syndrome / Volkmann	"Late sign is..." (pulselessness is LATE)	★★★★★
7	Bone tumors (osteosarcoma, Ewing's, metastasis, GCT)	"Onion-peel periosteal reaction" → Ewing's	★★★★★
8	Arthritis (OA, RA, gout, AS)	"Bamboo spine / HLA-B27"	★★★★★
9	Spine (disc prolapse, SLR, TB, cauda equina)	"Straight-leg raise positive at 30–60°"	★★★★★

Rank	Cluster	Typical stem style	Probability
10	Avascular necrosis (sites, causes, MRI)	"Most common cause of AVN of femoral head"	★★★★★

3. Stem-style trend (what examiners love)

- "All of the following are true EXCEPT" — tests the one outlier.
- "The most common / earliest / latest / best" — superlative stems.
- "The following investigation is the investigation of choice" — MRI for AVN/disc; X-ray first for trauma; USG for DDH <6 mo.
- "First-line management" — e.g., Ponseti for CTEV, Pavlik for DDH, debridement + ABx for open fracture.

4. Classic distractor traps (memorise to avoid)

- Pulselessness = **LATE** sign of compartment syndrome (pain + pain on passive stretch are early).
- Proximal scaphoid # AVN affects **proximal pole**, not waist.
- Posteromedial (posterior) hip dislocation injures **sciatic**, not femoral nerve.
- Fat embolism classically after **long-bone** (femur/tibia) fractures, not skull.
- Ewing's = **diaphysis**, osteosarcoma = **metaphysis**.
- RA cervical **subluxation** is a pre-op airway risk; not OA.
- In DDH, **Ortolani** = **reducible click**, Barlow = dislocatable.

5. Year-on-year direction (2024 → 2027)

- ↑ Image-based / X-ray interpretation.
- ↑ Clinical-scenario SBAs over pure fact recall.
- ↑ Bangladesh-epidemiology items (spinal TB, trauma, vitamin-D).
- → Stable core: classifications, nerve injuries, pediatric hip, tumors, arthritis.

6. How to exploit the trend

1. Build a **classification flash-card deck** (one card per classification).
2. Drill the "**most common / EXCEPT**" stem library daily.
3. Do **one X-ray-per-day** (Colle's, NOF, supracondylar, scaphoid, scoliosis, TB spine).
4. Convert every topic in `04-TOP-100-INDEX.md` into a 1-line SBA in your own words.

TOP 100 SURESHOT TOPICS — MASTER INDEX

Master Diploma Orthopaedics Admission Prep · Bangladesh 2027 · Probability-ranked

Legend: ★★★★★ Very High · ★★★★ High · ★★★ Moderate · ★★ Low · ★ Rare

Format per topic (in Parts 1–4): Notes · Rapid Revision · SBA/MCQ · Explanation · Trend · Clinical · Examiner Tip · Reference.

PART 1 — BASIC SCIENCES (25 topics)

Anatomy (8)

#	Topic	★	Reference
1	Brachial plexus (roots→trunks→cords→branches; Erb's C5–C6, Klumpke C8–T1)	★★★★★	Gray's / Snell
2	Sciatic & common peroneal (fibular) nerve; foot drop	★★★★★	Gray's / Snell
3	Femoral triangle & canal; saphenous opening (fossa ovalis)	★★★★	Gray's
4	Popliteal fossa — contents (popliteal artery/vein/tibial & common peroneal n.)	★★★★	Gray's
5	Carpal tunnel — median nerve; thenar muscles; Phalen/Tinel	★★★★★	Gray's / Snell
6	Hip joint blood supply — medial circumflex femoral, retinacular vessels, neck	★★★★★	Gray's
7	Knee joint — cruciates, menisci, locking/unlocking, nerve (saphenous/infrapatellar)	★★★★★	Gray's
8	Shoulder joint — stability, rotator cuff, axillary nerve (deltoid/surgical neck)	★★★★	Gray's / Snell

Physiology (5)

#	Topic	★	Reference
9	Skeletal muscle contraction — sliding filament, Ca ²⁺ , excitation–contraction coupling	★★★★	Guyton / Ganong
10	Calcium homeostasis — PTH, calcitonin, vit-D/calcitriol	★★★★★	Guyton / Ganong
11	Bone remodelling — osteoblast/osteoclast, RANKL/OPG, Wolff's law	★★★★	Guyton
12	Nerve AP & conduction velocity; demyelination effect	★★★	Ganong
13	Cartilage nutrition — diffusion from synovium; no vessels	★★★	Guyton

Biochemistry (3)

#	Topic	★	Reference
14	Collagen types & synthesis; osteogenesis imperfecta (type I)	★★★★★	Lippincott / Robbins
15	Vitamin-D metabolism; rickets vs osteomalacia	★★★★★	Lippincott
16	Gout — urate metabolism, xanthine oxidase, allopurinol/febuxostat	★★★★	Lippincott

Pathology (4)

#	Topic	★	Reference
17	Fracture healing — stages, callus, delayed/non-union factors	★★★★★	Robbins

#	Topic	★	Reference
18	Osteomyelitis — acute vs chronic, Brodie's abscess, sequestrum/involucrum	★★★★★	Robbins
19	Bone tumors overview — osteosarcoma, chondrosarcoma, Ewing's, metastasis	★★★★★	Robbins
20	Avascular necrosis — sites, risk factors, pathology	★★★★	Robbins

Pharmacology (3)

#	Topic	★	Reference
21	NSAIDs — COX-1/COX-2, aspirin, GI/renal, Reye's	★★★★★	Katzung
22	Bisphosphonates — mechanism, osteonecrosis of jaw, atypical femur #	★★★★	Katzung
23	Local anaesthetics/spinal/epidural; tourniquet pharmacology	★★★	Katzung

Microbiology (2)

#	Topic	★	Reference
24	Ortho infections — S. aureus (MRSA), Salmonella in sickle cell, M. tuberculosis	★★★★★	Lippincott
25	Prophylactic antibiotics in ortho — cefazolin, cement, biofilm	★★★★	Lippincott

PART 2 — FACULTY SURGERY (20 topics)

#	Topic	★	Reference
26	Shock — classification, hypovolaemic, management	★★★★	Bailey & Love / Schwartz
27	Fluid/electrolyte — burns, dehydration, hyperkalaemia	★★★★	Schwartz
28	Wound healing — primary/secondary, factors, suturing	★★★★	Bailey & Love
29	Tetanus — prophylaxis, TIG, management	★★★★★	Bailey & Love
30	Gas gangrene — C. perfringens, debridement, HBOT	★★★★	Bailey & Love
31	Blood transfusion — reactions, indications, components	★★★	Sabiston
32	Polytrauma / ATLS — primary survey, C-spine, damage control	★★★★★	ATLS / Bailey
33	Compartment syndrome — 5 Ps, pressure, fasciotomy	★★★★★	Bailey & Love
34	Burns — rule of nines, Parkland, escharotomy	★★★★	Schwartz
35	Soft-tissue tumours — lipoma vs sarcoma, biopsy principle	★★★	Schwartz
36	Thyroid/parathyroid — calcium link to bone	★★★	Schwartz
37	Acute limb ischaemia — 5 Ps, Rutherford, embolus	★★★★	Rutherford / Bailey
38	DVT/PE — Virchow's triad, prophylaxis, anticoagulants	★★★★★	Bailey & Love
39	Amputation — levels, indications, phantom limb	★★★	Sabiston
40	Gait & limp — Trendelenburg, antalgic, Trendelenburg test	★★★★★	Apley

#	Topic	★	Reference
41	Peripheral nerve injury — Seddon/Sunderland, neurapraxia	★★★★★	Bailey & Love
42	Hand infections — felon, paronychia, flexor tenosynovitis (Kanavel)	★★★★	Bailey & Love
43	Skin flaps & grafts — split/full-thickness, myocutaneous flap	★★★	Schwartz
44	Triage & disaster — mass-casualty categories	★★	ATLS
45	Sepsis & SIRS — definitions, qSOFA	★★★★	Surviving Sepsis

PART 3 — ORTHOPAEDICS HIGH-YIELD (45 topics)

Fractures & trauma (15)

#	Topic	★	Reference
46	Femoral neck # — Garden I–IV, blood supply, AVN/nonunion	★★★★★	Apley / Rockwood
47	Intertrochanteric # — Evans, extramedullary (DHS)	★★★★★	Campbell
48	Femoral shaft # — Winquist, IM nail	★★★★★	Rockwood
49	Supracondylar humerus # — Gartland, neurovascular (ANT interosseous/brachial a.)	★★★★★	Apley
50	Colle's # — dinner-fork, management	★★★★★	Apley
51	Scaphoid # — anatomical snuffbox, proximal-pole AVN	★★★★★	Apley
52	Tibial plateau # — Schatzker I–VI	★★★★	Schatzker / Rockwood
53	Ankle # — Weber (A/B/C) & Lauge-Hansen	★★★★★	Rockwood
54	Calcaneal # — Böhler's angle, Sanders	★★★★	Rockwood
55	Hip dislocation — posterior 90%, sciatic nerve	★★★★★	Apley
56	Pelvic # — Tile, APC/LC, haemodynamics	★★★★	Rockwood
57	Open # — Gustilo-Anderson I–III, debridement, timing	★★★★★	AO Principles
58	Paediatric # — greenstick, torus, Salter-Harris I–V	★★★★★	Apley
59	Fat embolism — long bone, petechiae, respiratory	★★★★★	Apley
60	DVT/PE in ortho — prophylaxis (pharmaco + mechanical)	★★★★	AAOS

Paediatric orthopaedics (10)

#	Topic	★	Reference
61	DDH — Barlow/Ortolani, Graf USG, Pavlik harness	★★★★★	Apley
62	CTEV (clubfoot) — Pirani, Ponseti casting	★★★★★	Apley
63	Cerebral palsy — GMFCS, spasticity, botulinum	★★★★	Campbell
64	Physeal growth & Salter-Harris implications	★★★	Apley
65	SCFE — Klein's line, pins-in-situ	★★★★★	Apley
66	Perthes — Waldenström, Herring/Lateral-pillar	★★★★★	Apley

#	Topic	★	Reference
67	Rickets/osteomalacia — widened epiphysis, bowing	★★★★★	Robbins
68	Osteogenesis imperfecta — blue sclerae, bisphosphonate	★★★★	Robbins
69	Achondroplasia & skeletal dysplasias	★★★★	Robbins
70	Congenital pseudoarthrosis tibia	★★★	Apley

Spine (8)

#	Topic	★	Reference
71	Cervical spine — cord syndromes (central, Brown-Séquard)	★★★★★	Apley
72	Lumbar disc prolapse — SLR, nerve root level	★★★★★	Apley
73	Spondylolisthesis — isthmic, Meyerding	★★★★	Apley
74	Scoliosis — Cobb angle, idiopathic, Milwaukee/brace	★★★★★	Apley
75	Kyphosis & Scheuermann's	★★★	Apley
76	Spinal TB (Pott's) — cold abscess, angular kyphosis	★★★★★	Apley / Robbins
77	Spinal stenosis — neurogenic claudication	★★★★	Apley
78	Cauda equina syndrome — red flag, emergency	★★★★★	Apley

Joints & arthritis (8)

#	Topic	★	Reference
79	Osteoarthritis — Kellgren-Lawrence, JSN	★★★★★	Apley
80	Rheumatoid arthritis — swan-neck/boutonnière, DMARDs, C-spine	★★★★★	Apley
81	Septic arthritis — hot joint, aspiration, Gram+	★★★★★	Apley / Robbins
82	Gout & pseudogout — birefringence, chondrocalcinosis	★★★★★	Robbins
83	Ankylosing spondylitis — bamboo spine, HLA-B27, Schober	★★★★★	Apley
84	AVN — femoral head, steroids, MRI (gold standard)	★★★★★	Apley
85	Charcot joint — neuropathic (diabetes)	★★★★	Apley
86	Hemophilic arthropathy	★★★★	Apley

Sports & soft tissue (4)

#	Topic	★	Reference
87	ACL/PCL — Lachman, pivot shift, drawer	★★★★★	Apley
88	Meniscal injury — McMurray, Apley, bucket-handle	★★★★★	Apley
89	Rotator cuff — supraspinatus, drop-arm, impingement	★★★★★	Apley
90	Tendinopathy — tennis/golfer elbow, Achilles	★★★★	Apley

Tumours, infection, misc (10)

#	Topic	★	Reference
91	Osteosarcoma — sunburst, Codman triangle, metaphysis	★★★★★	Robbins / Camp
92	Ewing's sarcoma — onion-peel, t(11;22), diaphysis	★★★★★	Robbins
93	Osteomyelitis (paediatric haematogenous) — femur metaphysis	★★★★★	Robbins
94	Bone metastasis — prostate (blastic), breast/lung (lytic)	★★★★★	Robbins
95	Giant-cell tumour — soap-bubble, epiphysis, Campanacci	★★★★★	Robbins
96	Soft-tissue sarcoma — staging, wide excision	★★★★	Schwartz
97	Polytrauma — ATLS, damage-control ortho	★★★★★	ATLS
98	Osteoporosis — DEXA T-score, bisphosphonate, falls	★★★★★	Apley
99	Imaging in ortho — X-ray views, MRI, bone scan, CT	★★★★★	Apley
100	Principles of fracture fixation — AO, absolute vs relative stability, ORIF/BO	★★★★★	AO Principles

PART 4 — TOP REPEATING COMMON QUESTIONS ("10000% COMMON") (10 topics)

#	Cross-cutting "most-repeated" fact	★
R1	Most common organism in osteomyelitis & septic arthritis → S. aureus	★★★★★
R2	Most common bone tumour in children → osteosarcoma ; in adults → metastasis	★★★★★
R3	Most common fracture in osteoporosis → vertebral > Colles' > hip (NOF)	★★★★★
R4	Most common nerve injured in humeral-shaft # → radial ; supracondylar → median (ANT interosseous)/brachial a.	★★★★★
R5	Most common cause of AVN → fracture neck femur / steroids / dislocation	★★★★★
R6	Most common site of paediatric haematogenous osteomyelitis → femur/distal femur metaphysis	★★★★★
R7	Most common joint replaced → knee > hip ; most common THR approach → posterior	★★★★
R8	Most common congenital ortho anomaly → CTEV (clubfoot) ; hip → DDH	★★★★
R9	Most common cause of limp in a child → transient synovitis (Irritable hip)	★★★★
R10	Most-repeated classifications → Garden, Gustilo, Salter-Harris, Weber, Gartland, Schatzker, Cobb, Evans	★★★★★

Next step (Turn 2): `05-BASIC-SCIENCES.md` — full notes + SBA/MCQ + explanations for topics #1–#25.

PART 1 — BASIC SCIENCES (Topics #1–#25)

High-Yield Notes · Rapid Revision · Repeating SBA/MCQ · Explanation · Trend · Clinical · Examiner Tip · Reference
Refs: Gray's Anatomy · Snell · Guyton & Hall · Ganong · Lippincott Biochem · Robbins · Katzung

1. Brachial Plexus ★★★★★ (Anatomy)

- **Notes:** Roots C5–T1 → trunks (upper C5–C6, middle C7, lower C8–T1) → divisions (ant/post) → cords (lateral/medial/posterior, named by relation to axillary artery) → branches.
- Erb's palsy (C5–C6): "waiter's tip" — adducted, internally rotated, extended elbow (obstetric traction / shoulder dystocia).
- Klumpke's (C8–T1): claw hand + Horner's (ptosis, miosis, anhidrosis) if T1/stellate involved.
- **Rapid Rev:** "Reach To Drink Cold Beer" = Roots→Trunks→Divisions→Cords→Branches. Posterior cord → radial, axillary, thoracodorsal, subscapular.
- **SBA:** A baby with "waiter's tip" posture after difficult delivery has lesion of: **a) C5–C6 (Erb's)**.
- **MCQ:** Klumpke palsy is associated with: **Horner's syndrome**.
- **Explanation:** Upper trunk (C5–C6) supplies deltoid/biceps → loss gives the characteristic posture.
- **Trend:** Repeated in diploma & residency; "waiter's tip" & Horner's are evergreen.
- **Clinical:** Obstetric brachial plexus palsy; also RTAs (traction).
- **Examiner Tip:** "Waiter's tip" = Erb's (upper), NOT Klumpke. Horner's = lower (Klumpke).
- **Ref:** Gray's Anatomy, ch. Upper limb innervation.

2. Sciatic & Common Peroneal (Fibular) Nerve ★★★★★ (Anatomy)

- **Notes:** Sciatic (L4–S3) = tibial + common peroneal; exits via greater sciatic foramen below piriformis, runs in posterior thigh. Common peroneal winds round neck of fibula → foot drop (dorsiflexion/eversion loss).
- **Rapid Rev:** Hip posterior dislocation & # neck femur → sciatic (often common peroneal branch). Total knee replacement → common peroneal palsy.
- **SBA:** Foot drop after a fibular-neck injury is due to lesion of: **common peroneal nerve**.
- **MCQ:** Sciatic nerve is formed by: **L4, L5, S1, S2, S3**.
- **Explanation:** Common peroneal is superficial at fibular neck → vulnerable to traction/casts.
- **Trend:** Very high — nerve-injury stem favourite.
- **Clinical:** Knee replacements, sciatic nerve block, comet-tail palsy.
- **Examiner Tip:** Foot drop = common peroneal; not tibial (tibial → plantarflexion loss, "foot slap"/calcaneal gait).
- **Ref:** Gray's, lower limb.

3. Femoral Triangle & Canal ★★★★★ (Anatomy)

- **Notes:** Inguinal lig (superior), sartorius (lateral), adductor longus (medial). Contents (lat→med): femoral nerve, artery, vein, empty canal, lymph (NAVEL mnemonic). Femoral canal = medial, site of femoral hernia. Saphenous opening (fossa ovalis) in fascia lata — saphenous vein pierces it.
- **Rapid Rev:** NAVEL = Nerve, Artery, Vein, Empty, Lymph. Femoral hernia = below inguinal lig, medial to pubic tubercle.
- **SBA:** Most medial structure in femoral canal: **lymph node / empty space**; femoral hernia arises here.
- **MCQ:** Femoral hernia is common in: **females** (wider pelvis).
- **Explanation:** Femoral canal is the only weak point → hernia; risk of strangulation.

- **Trend:** High; anatomy-of-groin question.
- **Clinical:** Femoral hernia vs inguinal hernia differentiation.
- **Examiner Tip:** Femoral hernia = inferolateral to pubic tubercle; inguinal = superomedial.
- **Ref:** Gray's, lower limb.

4. Popliteal Fossa ★★★★★ (Anatomy)

- **Notes:** Diamond space behind knee. Contents (superficial→deep): tibial nerve, popliteal vein, popliteal artery (TVA). Popliteal artery most vulnerable to aneurysm / entrapment.
- **Rapid Rev:** "TVA" order from superficial to deep: Tibial nerve, Vein, Artery.
- **SBA:** Deepest structure in popliteal fossa: **popliteal artery**.
- **MCQ:** Popliteal aneurysm most commonly affects: **popliteal artery**.
- **Explanation:** Artery lies deepest, against tibia → protected but dangerous if injured.
- **Trend:** Moderate-high; artery/nerve relation question.
- **Clinical:** Baker's cyst, popliteal aneurysm, soft-tissue tumours.
- **Examiner Tip:** Artery is DEEPEST (not vein).
- **Ref:** Gray's.

5. Carpal Tunnel ★★★★★ (Anatomy)

- **Notes:** Osteofibrous tunnel at wrist; median nerve + 9 flexor tendons (FPL, 4 FDS, 4 FDP). Thenar muscles (thenar eminence) = median (recurrent branch) except adductor pollicis (ulnar). Tests: Phalen (full wrist flexion 60 s), Tinel (tap median nerve).
- **Rapid Rev:** Median nerve = LOAF (Lumbricals 1–2, Opponens pollicis, Abductor pollicis brevis, Flexor pollicis brevis). "Ape hand" = median (thenar wasting).
- **SBA:** Classic finding in carpal tunnel: **thenar wasting + paraesthesia in thumb/index/middle**.
- **MCQ:** Median nerve supplies all EXCEPT: **adductor pollicis (ulnar)**.
- **Explanation:** Recurrent branch of median supplies thenar; compression → ape-hand/weak pinch.
- **Trend:** Very high; "EXCEPT" and muscle-supply stems.
- **Clinical:** Pregnancy, RA, hypothyroidism, repetitive strain.
- **Examiner Tip:** Adductor pollicis = ULNAR (not median) — classic trap.
- **Ref:** Gray's / Snell, hand.

6. Hip Joint Blood Supply ★★★★★ (Anatomy)

- **Notes:** Medial circumflex femoral artery (MCFA) → retinacular branches (posterior superior/inferior) = main intra-capsular blood to femoral head. foveal artery (ligamentum teres) minor in adults. Intertrochanteric (extracapsular) has rich anastomosis → heals; neck (intracapsular) precarious → AVN.
- **Rapid Rev:** MCFA = "Medial = Most important." Intracapsular # → retinacular vessels torn → AVN/nonunion.
- **SBA:** Main blood supply to femoral head: **medial circumflex femoral artery (retinacular vessels)**.
- **MCQ:** Fracture with highest AVN risk: **intracapsular femoral neck fracture**.
- **Explanation:** Retinacular vessels run under capsule; displaced neck # disrupts them → AVN.
- **Trend:** Very high; links anatomy → AVN.
- **Clinical:** Explains Garden-type nonunion/AVN.
- **Examiner Tip:** MCFA, not femoral artery, not foveal.
- **Ref:** Gray's, hip.

7. Knee Joint ★★★★★ (Anatomy)

- **Notes:** Tibiofemoral + patellofemoral. ACL (ant drawer, Lachman) prevents ant tibial translation; PCL (post drawer) prevents post translation. Menisci = medial (C, firm) / lateral (O). Locking (screw-home, medial rotation of femur) needs popliteus to unlock. Nerve: infrapatellar branch of saphenous + articular branches.
- **Rapid Rev:** ACL = ant, "turn & pivot"; PCL = post (dashboard). Locking = medial femoral rotation; unlocking = popliteus.
- **SBA:** Unlocking of knee is by: **popliteus muscle**.
- **MCQ:** Medial meniscus is more commonly injured because: **it is fixed to tibial collateral ligament**.
- **Explanation:** Medial meniscus attached to MCL → less mobile → torn more.
- **Trend:** Very high; cruciate/meniscus + popliteus.
- **Clinical:** ACL reconstruction, meniscectomy.
- **Examiner Tip:** Locking = medial rotation; popliteus UNLOCKS.
- **Ref:** Gray's, knee.

8. Shoulder Joint ★★★★★ (Anatomy)

- **Notes:** Ball-and-socket, most mobile, least stable. Rotator cuff = SITS (Supraspinatus, Infraspinatus, Teres minor, Subscapularis) — holds head in glenoid. Axillary nerve (C5–C6) winds round surgical neck → deltoid loss ("square shoulder"), loss of abduction first 15°.
- **Rapid Rev:** "SITS": Supraspinatus (abducts 0–15), Infraspinatus (ext rot), Teres minor (ext rot), Subscapularis (int rot). Axillary n. = surgical neck #.
- **SBA:** Nerve injured in surgical neck # humerus: **axillary nerve (C5–C6)**.
- **MCQ:** First 15° abduction is by: **supraspinatus**.
- **Explanation:** Supraspinatus initiates abduction; deltoid continues but needs cuff for stability.
- **Trend:** High; nerve + cuff.
- **Clinical:** Fracture neck humerus, shoulder dislocation (axillary n.).
- **Examiner Tip:** Axillary = surgical neck; radial = spiral groove; not the same.
- **Ref:** Gray's / Snell.

9. Skeletal Muscle Contraction ★★★★★ (Physiology)

- **Notes:** Sliding filament: Ca^{2+} binds troponin-C → tropomyosin shifts → myosin cross-bridges bind actin → power stroke. Excitation–contraction coupling: AP → T-tubule → RyR releases SR Ca^{2+} . Relaxation = Ca^{2+} pumped back (SERCA), ATP needed.
- **Rapid Rev:** " Ca^{2+} is the trigger." Rigor mortis = ATP depletion, cross-bridges stuck.
- **SBA:** Calcium binds to: **troponin C**.
- **MCQ:** Rigor mortis is due to: **loss of ATP → persistent actin-myosin binding**.
- **Explanation:** Without ATP, myosin cannot detach → stiffness.
- **Trend:** High; basic physio favourite.
- **Clinical:** Malignant hyperthermia (RYR1), neuroleptic malignant syndrome.
- **Examiner Tip:** Troponin C (not tropomyosin, not actin).
- **Ref:** Guyton ch. Muscle; Ganong.

10. Calcium Homeostasis ★★★★★ (Physiology)

- **Notes:** PTH ↑ Ca^{2+} (bone resorption, renal reabsorption, ↑ 1α -hydroxylase → calcitriol). Calcitriol ($1,25\text{-OH vit D}$) ↑ gut $\text{Ca}^{2+}/\text{PO}_4$ absorption. Calcitonin (C-cells, thyroid) ↓ Ca^{2+} . Normal Ca^{2+} ~8.5–10.5 mg/dL; ionised is active.
- **Rapid Rev:** PTH = "bone kidney gut" ↑ Ca; Calcitonin opposes; Vit D = gut absorption.
- **SBA:** Most important regulator of plasma Ca^{2+} : **PTH**.

- **MCQ:** 1,25-dihydroxyvitamin D is formed in: **kidney (proximal tubule)**.
- **Explanation:** Liver 25-hydroxylation, kidney 1 α -hydroxylation.
- **Trend:** Very high; bone-metabolism link.
- **Clinical:** Hypoparathyroidism (Tetany, Chvostek/Trousseau), hyperCa (stones/bones/groans).
- **Examiner Tip:** Vitamin D activated in KIDNEY, not skin, not liver final step.
- **Ref:** Guyton / Ganong, endocrine.

11. Bone Remodelling ★★★★★ (Physiology)

- **Notes:** Osteoblasts form bone (osteoid, type I collagen); osteoclasts resorb (multinucleate, RANK-L from osteoblast $\rightarrow$ RANK on osteoclast). OPG inhibits RANKL. Wolff's law: bone adapts to stress. Coupling = remodelling units.
- **Rapid Rev:** RANKL $\rightarrow$ RANK = resorption; OPG = brake. Wolff = form follows function.
- **SBA:** Osteoclast formation is stimulated by: **RANKL**.
- **MCQ:** Wolff's law states: **bone architecture remodels according to mechanical stress**.
- **Explanation:** Mechanical loading directs remodelling; immobilisation $\rightarrow$ resorption.
- **Trend:** High; osteoporosis/bisphosphonate link.
- **Clinical:** Osteoporosis (anti-RANKL denosumab), immobilisation osteoporosis.
- **Examiner Tip:** OPG = antagonist of RANKL.
- **Ref:** Guyton.

12. Nerve Action Potential & Conduction ★★★ (Physiology)

- **Notes:** Depolarisation = Na $^{+}$ influx (voltage-gated Na channels); repolarisation = K $^{+}$ efflux. Myelinated fibres saltatory conduction (nodes of Ranvier) $\rightarrow$ faster. Demyelination (MS, GBS) $\rightarrow$ slowed/blocked conduction, $\downarrow$ velocity.
- **Rapid Rev:** Saltatory = jumping node $\rightarrow$ node. Demyelination = $\downarrow$ velocity, not $\downarrow$ amplitude primarily.
- **SBA:** Saltatory conduction occurs in: **myelinated nerve fibres**.
- **MCQ:** In demyelination, what is primarily reduced: **conduction velocity**.
- **Explanation:** Saltatory conduction lost $\rightarrow$ slow impulse.
- **Trend:** Moderate; basic neurophysio.
- **Clinical:** Guillain-Barré, MS.
- **Examiner Tip:** Demyelination $\rightarrow$ velocity $\downarrow$ (amplitude may be preserved early).
- **Ref:** Ganong.

13. Cartilage Nutrition ★★★ (Physiology)

- **Notes:** Articular cartilage is avascular, aneural, alymphatic. Nutrition by diffusion from synovial fluid during joint loading/unloading (pump mechanism). Chondrocytes in lacunae, matrix = type II collagen + proteoglycan (aggrecan, GAGs).
- **Rapid Rev:** "Cartilage feeds by diffusion from synovium." No vessels = poor healing.
- **SBA:** Articular cartilage receives nutrition mainly from: **synovial fluid (diffusion)**.
- **MCQ:** Cartilage matrix predominantly contains: **type II collagen + proteoglycans**.
- **Explanation:** Avascular $\rightarrow$ limited repair; basis of OA.
- **Trend:** Moderate; OA link.
- **Clinical:** OA (cartilage loss), cartilage injuries.
- **Examiner Tip:** Type II (cartilage), Type I (bone/tendon/skin).
- **Ref:** Guyton.

14. Collagen Types & Synthesis ★★★★★ (Biochemistry)

- **Notes:** Type I (bone, tendon, skin, cornea) — deficiency = Osteogenesis imperfecta (brittle bones, blue sclerae). Type II (cartilage). Type III (blood vessels, granulation, skin — Ehlers-Danlos/vascular). Type IV (basement membrane). Synthesis: procollagen → procollagen (hydroxyproline/hydroxylysine need Vit C = prolyl/lysyl hydroxylase) → triple helix → secretion → cleavage → fibrils.
- **Rapid Rev:** "Be (I) a BONE, Tendon, Skin." Vit C needed for hydroxylation; deficiency = scurvy.
- **SBA:** Osteogenesis imperfecta is due to defect in: **type I collagen**.
- **MCQ:** Hydroxylation of proline/lysine in collagen requires: **vitamin C (ascorbate)**.
- **Explanation:** No Vit C → defective collagen → scurvy (bleeding gums, poor healing).
- **Trend:** Very high; OI + scurvy.
- **Clinical:** OI, scurvy, Ehlers-Danlos.
- **Examiner Tip:** OI = Type I (not II). Scurvy = no Vit C hydroxylation.
- **Ref:** Lippincott Biochem; Robbins.

15. Vitamin D Metabolism ★★★★★ (Biochemistry)

- **Notes:** Skin 7-dehydrocholesterol → cholecalciferol (D3) via UV; diet D2/D3 → liver 25-OH (storage) → kidney 1 α -OH → 1,25-(OH) $_2$ D (active). Deficiency in children = rickets (wide epiphyses, rachitic rosary, bowing); adults = osteomalacia (bone pain, Looser's zones). Breastfed infants in BD need supplementation.
- **Rapid Rev:** Rickets = children (growth plate open); Osteomalacia = adults (defective mineralisation).
- **SBA:** Active form of vitamin D: **1,25-dihydroxycholecalciferol**.
- **MCQ:** Rickets is characterised by: **widened epiphyses & bowing of long bones**.
- **Explanation:** Unmineralised osteoid at growth plate → deformity.
- **Trend:** Very high; Bangladesh endemic.
- **Clinical:** Nutritional rickets (BD common), renal osteodystrophy.
- **Examiner Tip:** Rickets = children; Osteomalacia = adults (same deficiency).
- **Ref:** Lippincott.

16. Gout ★★★★★ (Biochemistry)

- **Notes:** Urate metabolism: purine breakdown → xanthine → uric acid (xanthine oxidase). Hyperuricaemia → monosodium urate crystals in joint (negatively birefringent, needle-shaped under polarised light). Allopurinol/febuxostat inhibit xanthine oxidase; probenecid ↑ excretion; colchicine/NSAIDs acute.
- **Rapid Rev:** Gout crystals = **NEGATIVELY** birefringent needles. Pseudogout (CPP) = positively birefringent rhomboids.
- **SBA:** Gout crystals under polarised light are: **negatively birefringent, needle-shaped**.
- **MCQ:** Xanthine oxidase inhibitor used in gout: **allopurinol**.
- **Explanation:** Allopurinol reduces uric acid production.
- **Trend:** High; crystal differentiation.
- **Clinical:** Podagra (1st MTP), tophi, renal stones.
- **Examiner Tip:** Gout = **NEGATIVE**; Pseudogout (CPPD) = **POSITIVE** birefringence.
- **Ref:** Lippincott; Robbins.

17. Fracture Healing ★★★★★ (Pathology)

- **Notes:** Stages: haematoma → inflammatory (week 1) → soft callus (fibrocartilage, weeks 2–3) → hard callus (woven bone, weeks 4–12) → remodelling (months–years, lamellar). Primary (direct) healing with rigid fixation (no callus). Delayed union >expected; non-union (no healing, pseudoarthrosis); malunion.

- **Rapid Rev:** Factors delaying: infection, poor blood, distraction, smoking, NSAIDs, radiotherapy, malnutrition.
- **SBA:** Hard callus is composed of: **woven bone**.
- **MCQ:** Which delays fracture healing: **infection / smoking / NSAIDs / distraction** (all EXCEPT good reduction).
- **Explanation:** Infection & instability are top causes of non-union.
- **Trend:** Very high; healing + non-union causes.
- **Clinical:** Atrophic vs hypertrophic non-union.
- **Examiner Tip:** Primary healing = NO callus (rigid compression plate).
- **Ref:** Robbins, bone repair.

18. Osteomyelitis ★★★★★ (Pathology)

- **Notes:** Acute haematogenous (children) → femur/distal femur metaphysis (slow flow, metaphyseal loops). *S. aureus* #1. Neonates: Group B strep, *E. coli*. Sick cell: *Salmonella*. Chronic → sequestrum (dead bone) + involucrum (new shell) + sinus. Brodie's abscess = subacute localised.
- **Rapid Rev:** "Metaphysis in kids." Sequestrum = dead bone; Involucrum = living shell.
- **SBA:** Most common organism in acute haematogenous osteomyelitis: ***S. aureus***.
- **MCQ:** In sickle-cell disease osteomyelitis is commonly due to: ***Salmonella***.
- **Explanation:** Vaso-occlusion + infarcted marrow → *Salmonella* favoured.
- **Trend:** Very high; organism + sequestrum.
- **Clinical:** Chronic osteomyelitis, diabetic foot osteo.
- **Examiner Tip:** Sequestrum = dead bone (not involucrum).
- **Ref:** Robbins; Lippincott Micro.

19. Bone Tumours Overview ★★★★★ (Pathology)

- **Notes:** Osteosarcoma = metaphysis, sunburst, Codman triangle, young. Chondrosarcoma = pelvis/shoulder, cartilage matrix, adults. Ewing's = diaphysis, onion-skin, t(11;22), <20 yr. Giant-cell = epiphysis, soap-bubble, 20–40 yr. Metastasis = most common malignant bone tumour (prostate = blastic, breast/lung/thyroid = lytic).
- **Rapid Rev:** Osteo = metaphysis/sunburst; Ewing = diaphysis/onion; GCT = epiphysis/soap-bubble.
- **SBA:** Onion-skin periosteal reaction suggests: **Ewing's sarcoma**.
- **MCQ:** Most common malignant bone tumour overall: **metastasis**.
- **Explanation:** Metastases outnumber primary malignancies.
- **Trend:** Very high; imaging sign → diagnosis.
- **Clinical:** "Coda triangle," lung mets in osteosarcoma.
- **Examiner Tip:** Osteosarcoma = METAPHYSIS; Ewing = DIAPHYSIS.
- **Ref:** Robbins, bone tumours.

20. Avascular Necrosis ★★★★★ (Pathology)

- **Notes:** Ischaemic death of bone marrow + trabeculae → collapse. Common sites: femoral head, scaphoid proximal pole, talus, femoral condyle. Causes: #/dislocation (NOF), steroids, alcohol, sickle cell, Caisson, Gaucher. MRI = earliest/sensitive.
- **Rapid Rev:** "ASEPTIC": Alcohol, Steroids, Trauma, Idiopathic, Caisson, Lupus/Legg, Sickle.
- **SBA:** Investigation of choice for early AVN femoral head: **MRI**.
- **MCQ:** Most common cause of AVN of femoral head: **fracture neck femur / steroids**.
- **Explanation:** Disruption of MCFA retinacular vessels.
- **Trend:** High; investigation-of-choice.
- **Clinical:** Steroid-induced AVN in lupus/transplant.

- **Examiner Tip:** X-ray normal early → MRI is gold standard.
- **Ref:** Robbins; Apley.

21. NSAIDs ★★★★★ (Pharmacology)

- **Notes:** Inhibit cyclo-oxygenase → ↓ prostaglandins. COX-1 (housekeeping: gastric mucosa, platelets, kidney) vs COX-2 (inflammation). Aspirin irreversibly inhibits platelet COX-1 (anti-platelet). GI bleed, renal impairment, Reye's syndrome (aspirin in children/viral). COX-2 inhibitors (celecoxib) spare GI but ↑ CV risk.
- **Rapid Rev:** Aspirin = irreversible; Reye's = aspirin + viral in kids; COX-2 = GI-safe, CV-risk.
- **SBA:** Reye's syndrome is associated with: **aspirin in children with viral illness.**
- **MCQ:** Aspirin's antiplatelet effect is due to: **irreversible COX-1 inhibition.**
- **Explanation:** Permanent platelet COX blockade until new platelets form (~7–10 d).
- **Trend:** Very high; Reye's + COX.
- **Clinical:** Post-op analgesia, VTE prophylaxis adjunct.
- **Examiner Tip:** Reye's = aspirin (not paracetamol).
- **Ref:** Katzung.

22. Bisphosphonates ★★★★★ (Pharmacology)

- **Notes:** Pyrophosphate analogues; bind bone hydroxyapatite, ingested by osteoclasts → apoptosis. Used in osteoporosis, Paget's, metastases (↓ skeletal events), hyperCa. Adverse: oesophageal irritation (take upright), osteonecrosis of jaw (ONJ), atypical subtrochanteric femur fractures, hypocalcaemia/AVN of jaw.
- **Rapid Rev:** "Alendronate weekly, upright, wait 30 min." ONJ + atypical femur # = rare but tested.
- **SBA:** Atypical femoral fracture is a known adverse effect of: **bisphosphonates.**
- **MCQ:** Bisphosphonates act by: **inhibiting osteoclast-mediated bone resorption.**
- **Explanation:** Toxic to osteoclasts → net bone gain.
- **Trend:** High; ONJ + atypical fracture (now in exams).
- **Clinical:** Osteoporosis, Paget's, myeloma bone disease.
- **Examiner Tip:** ONJ (not AVN of hip) is the classical oral bisphos complication.
- **Ref:** Katzung.

23. Local Anaesthetics / Spinal / Epidural ★★★★★ (Pharmacology)

- **Notes:** Block Na⁺ channels (use-dependent) → no AP. Esters (procaine, cocaine) metabolised by plasma cholinesterase; amides (lidocaine, bupivacaine, ropivacaine) by liver. Spinal = intrathecal (subarachnoid), single shot, sympathetic block first. Epidural = epidural space, catheter. Bupivacaine cardiotoxic.
- **Rapid Rev:** Esters = -caine with ONE i (procaine); Amides = TWO i (lidocaine). Spinal → hypotension (sympathetic).
- **SBA:** Most cardiotoxic local anaesthetic: **bupivacaine.**
- **MCQ:** Spinal anaesthesia causes hypotension due to: **sympathetic blockade.**
- **Explanation:** Sympathetic tone lost → vasodilatation → hypotension.
- **Trend:** Moderate; ortho OT relevance.
- **Clinical:** TIVA/regional for joint replacement; tourniquet.
- **Examiner Tip:** Tourniquet = ischaemia; LA toxicity = CNS then cardiac.
- **Ref:** Katzung.

24. Orthopaedic Infections (Microbiology) ★★★★★ (Microbiology)

- **Notes:** *S. aureus* (incl. MRSA) = #1 in osteomyelitis, septic arthritis, implant infection. Neonatal osteo: GBS, *E. coli*, *S. aureus*. Sickie cell: *Salmonella*. TB spine (Pott's) = *M. tuberculosis*. Biofilm on implants protects bacteria (needs debridement + spacer + ABx).
- **Rapid Rev:** "Staph rules ortho." Septic arthritis in sexually active = *N. gonorrhoeae*.
- **SBA:** Most common organism in septic arthritis: ***S. aureus***.
- **MCQ:** Implant-related infection is protected by: **biofilm**.
- **Explanation:** Biofilm = recalcitrant; requires hardware removal.
- **Trend:** Very high; organism + biofilm.
- **Clinical:** MRSA, diabetic foot, TB spine.
- **Examiner Tip:** Gonococcal septic arthritis = young sexually active (not Staph in that niche).
- **Ref:** Lippincott Microbiology.

25. Prophylactic Antibiotics in Ortho ★★★★★ (Microbiology)

- **Notes:** Single pre-incision dose cefazolin (cephalosporin) 1st gen covers MSSA; allergy → clindamycin/vancomycin (MRSA cover). Given within 60 min before incision. Impregnated bone cement (antibiotic-loaded, e.g., gentamicin/vancomycin) for revision/septic.
- **Rapid Rev:** Cefazolin 1g IV pre-op. Cement = local ABx delivery.
- **SBA:** Prophylactic antibiotic of choice for clean ortho implant surgery: **cefazolin**.
- **MCQ:** Antibiotic-loaded bone cement is used to: **deliver local high-dose ABx & prevent biofilm**.
- **Explanation:** Local elution beats systemic for biofilm prevention.
- **Trend:** High; OT protocol.
- **Clinical:** Joint replacement, ORIF.
- **Examiner Tip:** Cefazolin (not cephalexin oral) for surgical prophylaxis.
- **Ref:** Lippincott; AAOS guidelines.

Next (Turn 3): `06-FACULTY-SURGERY.md` — topics #26–#45.

PART 2 — FACULTY SURGERY (Topics #26–#45)

Format: Notes · Rapid Rev · SBA/MCQ · Explanation · Trend · Clinical · Examiner Tip · Reference

Refs: Bailey & Love · Schwartz · Sabiston · ATLS

26. Shock ★★★★★

- **Notes:** Hypovolaemic (haemorrhage/trauma) most common in ortho; cardiogenic; distributive (septic/anaphylactic/neurogenic); obstructive (tension PTx, Tamponade, PE). Class IV haemorrhage >40% blood volume → shock.
- **Rapid Rev:** "HIPPO" = Hypovolaemic, Cardiogenic, Distributive, Obstructive. Shock = inadequate tissue perfusion.
- **SBA:** Most common type of shock in trauma: **hypovolaemic**.
- **MCQ:** Class IV haemorrhage corresponds to blood loss: **>2000 mL (>40%)**.
- **Explanation:** Replace intravascular volume (crystalloid then blood).
- **Trend:** High; classification stems.
- **Clinical:** Polytrauma, pelvic/# femur haemorrhage.

- **Examiner Tip:** Neurogenic shock = bradycardia + warm (not cold) skin.
- **Ref:** Bailey & Love, ch. Shock.

27. Fluid & Electrolyte ★★★★★

- **Notes:** Burns Parkland = $4 \text{ mL} \times \text{kg} \times \% \text{TBSA}$ over 24 h, HALF in first 8 h. Hyperkalaemia ECG: peaked T, widened QRS, sine wave → arrhythmias; treat Ca^{2+} gluconate, insulin+Dextrose, nebulised salbutamol.
- **Rapid Rev:** Parkland: 4-2-1? No — $4 \text{ mL/kg/\%TBSA}$. HyperK = peaked T.
- **SBA:** Parkland formula fluid for 50 kg, 40% burn in first 8 h: **4000 mL** ($4 \times 50 \times 40 = 8000/24\text{h}$; half=4000 in 8h).
- **MCQ:** Earliest ECG sign of hyperkalaemia: **peaked T waves**.
- **Explanation:** K^{+} shifts resting membrane potential.
- **Trend:** High; numerical Parkland tested.
- **Clinical:** Burns, crush injury (rhabdo→hyperK).
- **Examiner Tip:** Half the calculated 24h volume is given in the FIRST 8 h from time of burn.
- **Ref:** Schwartz, Burns.

28. Wound Healing ★★★★★

- **Notes:** Primary (intentional, clean <8–12 h); secondary (granulation, open, e.g., pressure sore); tertiary/delayed primary (contaminated then closed). Impaired by infection, DM, steroids, ischaemia, malnutrition, smoking, radiation.
- **Rapid Rev:** Infection = #1 cause of delayed healing. Vitamin C/protein needed.
- **SBA:** Healing by secondary intention occurs in: **open/contaminated wounds (e.g., pressure ulcer)**.
- **MCQ:** Which most impairs healing: **infection** (vs clean closure, good nutrition).
- **Explanation:** Bacteria → persistent inflammation → no maturation.
- **Trend:** High; factors stems.
- **Clinical:** Diabetic foot, dehiscence laparotomy.
- **Examiner Tip:** "Secondary" = granulation fills; not primary suture.
- **Ref:** Bailey & Love.

29. Tetanus ★★★★★

- **Notes:** Clostridium tetani spore; tetanospasmin blocks inhibitory interneurons (GABA/glycine) → sustained muscle spasm/risus sardonicus, opisthotonos, trismus (lockjaw). Prophylaxis: active TT + passive TIG for dirty wounds. Rx: wound debridement, TIG, metronidazole, benzodiazepines, airway.
- **Rapid Rev:** Toxin blocks INHIBITION → spasm. TIG = passive (immediate), TT = active.
- **SBA:** Tetanus toxin acts by: **blocking inhibitory neurotransmitters (GABA/glycine)**.
- **MCQ:** Passive immunisation in tetanus: **Tetanus Immunoglobulin (TIG)**.
- **Explanation:** TIG neutralises unbound toxin; cannot reverse bound toxin.
- **Trend:** Very high; mechanism + prophylaxis.
- **Clinical:** Road-traffic/open wounds (BD common).
- **Examiner Tip:** TIG (human) not ATS if available; "lockjaw" = trismus.
- **Ref:** Bailey & Love; Lippincott Micro.

30. Gas Gangrene ★★★★★

- **Notes:** Clostridium perfringens (α -toxin phospholipase C) → myonecrosis, crepitus, foul serosanguinous discharge, systemic toxicity, haemolysis. Rx: urgent wide debridement/amputation + penicillin+clindamycin + hyperbaric O₂.
- **Rapid Rev:** Crepitus + foul smell = gas gangrene. HBOT adjunct.

- **SBA:** Causative organism of gas gangrene: **Clostridium perfringens**.
- **MCQ:** Crepitus in a traumatic wound suggests: **gas-forming infection (C. perfringens)**.
- **Explanation:** Gas from fermentation in devitalised muscle.
- **Trend:** High; organism + crepitus.
- **Clinical:** War/RTA crush injuries.
- **Examiner Tip:** Gas gangrene = muscle (myonecrosis), not just skin.
- **Ref:** Bailey & Love.

31. Blood Transfusion ★★★

- **Notes:** Indication stable Hb <7–8 g/dL. Reactions: acute haemolytic (ABO, fever+hemoglobinuria, DIC), febrile non-haemolytic (most common), allergic, TRALI (donor anti-leukocyte → pulmonary oedema), TACO (volume), GVHD (irradiated blood for immunocompromised).
- **Rapid Rev:** TRALI = donor antibodies; TACO = volume overload.
- **SBA:** Most common transfusion reaction: **febrile non-haemolytic**.
- **MCQ:** TRALI is caused by: **donor anti-leukocyte antibodies**.
- **Explanation:** Neutrophil priming → pulmonary capillary leak.
- **Trend:** Moderate; reaction identification.
- **Clinical:** Polytrauma, major ortho (spine).
- **Examiner Tip:** TRALI = donor; TACO = recipient volume.
- **Ref:** Sabiston.

32. Polytrauma / ATLS ★★★★★

- **Notes:** Primary survey ABCDE; C-spine immobilisation; oxygenation; control haemorrhage; assess disability (GCS); exposure. Damage-control surgery for unstable. "Treat the greatest threat to life first."
- **Rapid Rev:** A irway+C-spine, B reathing, C irculation, D isability, E xposure.
- **SBA:** First step in ATLS primary survey: **airway with cervical-spine control**.
- **MCQ:** In polytrauma, priority is: **life-threatening airway/breathing/bleeding first**.
- **Explanation:** Algorithm = resuscitation before definitive fixation.
- **Trend:** Very high; trauma stems.
- **Clinical:** RTA, fall from height.
- **Examiner Tip:** Pelvic binder for unstable pelvic # before X-ray.
- **Ref:** ATLS Manual.

33. Compartment Syndrome ★★★★★

- **Notes:** ↑ interstitial pressure > perfusion → ischaemia. 5 Ps: Pain (disproportionate), Pain on passive stretch (EARLY), Paraesthesia, Pallor, Pulselessness & Paralysis (LATE). Dx: clinical + compartment pressure >30 mmHg. Rx: urgent fasciotomy.
- **Rapid Rev:** Pain + passive stretch = EARLY. Pulselessness = LATE (don't wait!).
- **SBA:** Earliest reliable sign of compartment syndrome: **pain on passive stretch**.
- **MCQ:** Late sign of compartment syndrome: **pulselessness / paralysis**.
- **Explanation:** Nerves/muscle ischaemia precede pulse loss.
- **Trend:** Very high; classic trap (pulselessness is late).
- **Clinical:** Tibial #, forearm (Volkmann), casts too tight, reperfusion.
- **Examiner Tip:** Absent pulse does NOT mean early — it's late.
- **Ref:** Bailey & Love; Apley.

34. Burns ★★★★★

- **Notes:** Rule of nines (adult): head 9, each arm 9, ant trunk 18, post trunk 18, each leg 18, perineum 1. Lund-Browder for children. Parkland resuscitation. Circumferential full-thickness → escharotomy for perfusion/ventilation.
- **Rapid Rev:** Adult: 9-18-18-18-18-1. Child: head bigger (18), leg smaller.
- **SBA:** Total body surface in rule of nines for both legs (adult): **36%**.
- **MCQ:** Indication for escharotomy: **circumferential full-thickness burn compromising circulation**.
- **Explanation:** Eschar acts as tourniquet.
- **Trend:** High; rule-of-nines numerical.
- **Clinical:** Electrical/flame burns.
- **Examiner Tip:** Use Lund-Browder in children (rule of nines overestimates head).
- **Ref:** Schwartz, Burns.

35. Soft-Tissue Tumours ★★★★★

- **Notes:** Lipoma = benign, soft, mobile. Sarcoma = deep, >5 cm, growing, painful, fixed; staging by size/depth/grade/metastasis. Biopsy principle: incisional, longitudinal, do NOT cross compartments, plan with surgeon.
- **Rapid Rev:** "Deep + >5 cm + painful" → suspect sarcoma.
- **SBA:** Suspicious soft-tissue mass should be biopsied by: **incisional biopsy planned with the treating surgeon**.
- **MCQ:** Benign soft-tissue tumour most common: **lipoma**.
- **Explanation:** Bad biopsy can upstage/worsen resection.
- **Trend:** Moderate; biopsy principle tested.
- **Clinical:** Extremity masses.
- **Examiner Tip:** NEVER excise a suspected sarcoma widely without biopsy plan.
- **Ref:** Schwartz; Camp.

36. Thyroid / Parathyroid ★★★★★

- **Notes:** Primary hyperparathyroidism → ↑ PTH → hypercalcaemia (stones, bones, groans, psychiatric moans), brown tumours of bone (osteoclastoma-like), osteoporosis. #1 cause = solitary adenoma.
- **Rapid Rev:** Brown tumour = giant-cell reparative granuloma in hyperPTH.
- **SBA:** Brown tumour of bone is seen in: **hyperparathyroidism**.
- **MCQ:** 'Stones, bones, groans' describe: **hypercalcaemia of hyperparathyroidism**.
- **Explanation:** PTH ↑ bone resorption → cysts/brown tumours.
- **Trend:** Moderate; bone link.
- **Clinical:** Renal stones, pathological #.
- **Examiner Tip:** Brown tumour = NOT giant-cell tumour of tendon sheath.
- **Ref:** Schwartz, Endocrine.

37. Acute Limb Ischaemia ★★★★★

- **Notes:** 6 Ps: Pain, Pulseless, Pallor, Paraesthesia, Paralysis, Perishing cold. Rutherford categories (I viable, IIa/b threatened, III irreversible). Embolus (AF, saddle) vs in-situ thrombosis. Rx: heparin, embolectomy/thrombectomy.
- **Rapid Rev:** 6 Ps (add Perishing cold). Embolus = sudden, AF source.
- **SBA:** Sudden painful pulseless leg in AF: **acute arterial embolus**.
- **MCQ:** Most common source of arterial embolus: **cardiac (atrial fibrillation)**.

- **Explanation:** Embolus lodges at bifurcation (common femoral/popliteal).
- **Trend:** High; 6 Ps.
- **Clinical:** Acute popliteal embolism.
- **Examiner Tip:** Paralysis = irreversible if late (Rutherford III).
- **Ref:** Rutherford; Bailey & Love.

38. DVT / PE ★★★★★

- **Notes:** Virchow's triad: stasis, endothelial injury, hypercoagulability. Prophylaxis: LMWH (enoxaparin), mechanical (stockings, IPC), early mobilisation. Rx: anticoagulation; PE = chest pain, dyspnoea, hypoxia, Wells score.
- **Rapid Rev:** Ortho # (esp. hip/knee, pelvis) = high DVT risk. LMWH post-op.
- **SBA:** Prophylaxis of DVT after ortho surgery: **LMWH + early mobilisation + compression.**
- **MCQ:** Virchow's triad includes: **stasis, endothelial injury, hypercoagulability.**
- **Explanation:** Immobility + surgery → stasis + hypercoagulability.
- **Trend:** Very high; prophylaxis stems.
- **Clinical:** THR/TKR, pelvis/# femur.
- **Examiner Tip:** LMWH, not aspirin, is standard VTE prophylaxis.
- **Ref:** Bailey & Love; AAOS.

39. Amputation ★★★

- **Notes:** Indications: ischaemia (PVD), trauma (unsalvageable), tumour, infection (gas gangrene), congenital. Levels: above/below knee, Syme's, through-knee. Phantom limb pain common. Myoplastic closure.
- **Rapid Rev:** Below-knee (BK) better functional outcome than AK.
- **SBA:** Most common level of lower-limb amputation: **below-knee (transtibial).**
- **MCQ:** Phantom limb sensation is: **common and normal post-amputation.**
- **Explanation:** Cortical remap → perceived sensation.
- **Trend:** Moderate; levels.
- **Clinical:** Diabetic foot, vascular.
- **Examiner Tip:** BK preserves knee → better prosthesis function.
- **Ref:** Sabiston.

40. Gait & Limp ★★★★★

- **Notes:** Trendelenburg (gluteus medius weakness → contralateral pelvic drop; test: stand on one leg). Antalgic (pain → short stance). Short-leg (circumduction). Spastic (hemiplegic). Waddling (proximal myopathy/DMD).
- **Rapid Rev:** Trendelenburg +ve = pelvic drop on unsupported side.
- **SBA:** Positive Trendelenburg test indicates weakness of: **gluteus medius.**
- **MCQ:** Limp with pain relieved by rest = **antalgic limp.**
- **Explanation:** Gluteus medius stabilises pelvis in single-leg stance.
- **Trend:** Very high; gait test.
- **Clinical:** DDH, Perthes, hip OA, polio.
- **Examiner Tip:** Trendelenburg test = STAND on affected leg, pelvis drops on normal side.
- **Ref:** Apley.

41. Peripheral Nerve Injury ★★★★★

- **Notes:** Seddon: neurapraxia (conduction block, recovers), axonotmesis (axon interrupted, Wallerian, regenerates 1 mm/day), neurotmesis (transection). Sunderland 1–5. Tinel = regenerating nerve (tingling distal to percussion).
- **Rapid Rev:** Neurapraxia = temporary; neurotmesis = needs surgery.
- **SBA:** Conduction block without axonal disruption: **neurapraxia**.
- **MCQ:** Rate of peripheral nerve regeneration: **~1 mm/day**.
- **Explanation:** Axonotmesis regenerates along intact endoneurium.
- **Trend:** Very high; classification.
- **Clinical:** Saturday-night palsy (radial), bracelet injury (ulnar).
- **Examiner Tip:** Neurapraxia recovers fully (e.g., transient foot drop).
- **Ref:** Bailey & Love.

42. Hand Infections ★★★★★

- **Notes:** Felon (pulp space), paronychia (nail fold), flexor tenosynovitis (Kanavel: finger held flexed, uniform swelling, tenderness along sheath, pain on extension), thenar/web-space abscess.
- **Rapid Rev:** Kanavel's 4 signs = tenosynovitis.
- **SBA:** Flexor tenosynovitis is suggested by: **Kanavel's signs (flexed posture + tenderness along tendon)**.
- **MCQ:** Most common hand infection: **paronychia**.
- **Explanation:** Sheath infection can destroy tendon rapidly.
- **Trend:** High; Kanavel signs.
- **Clinical:** Diabetic, penetrating injury.
- **Examiner Tip:** Flexor tenosynovitis = surgical emergency (not just antibiotics).
- **Ref:** Bailey & Love.

43. Skin Flaps & Grafts ★★★★★

- **Notes:** Split-thickness graft (epidermis + partial dermis; donor = thigh; meshed). Full-thickness (better cosmesis; small defects). Myocutaneous/rotational flap = vascularised tissue for coverage (e.g., trauma, pressure sore, tumour).
- **Rapid Rev:** STSG takes on granulating bed; flap carries its own blood supply.
- **SBA:** Split-thickness skin graft includes: **epidermis + part of dermis**.
- **MCQ:** Flap differs from graft in having: **its own blood supply**.
- **Explanation:** Grafts rely on recipient bed; flaps are pedicled/free.
- **Trend:** Moderate; flap vs graft.
- **Clinical:** Trauma coverage, decubitus.
- **Examiner Tip:** Free flap = microvascular anastomosis.
- **Ref:** Schwartz, Plastic.

44. Triage & Disaster ★★

- **Notes:** Simple triage: Red = immediate (life-threatening, salvageable), Yellow = delayed, Green = minor (walking wounded), Black = deceased/expectant.
- **Rapid Rev:** START: Resp >30 / nil → red/black; perfusion/mental → sort.
- **SBA:** Immobile breathing patient after disaster with no radial pulse: **red (immediate)** or black if apnoeic after airway.
- **MCQ:** Walking wounded are tagged: **green**.
- **Explanation:** Prioritise salvageable life-threatening.
- **Trend:** Low; mass-casualty.
- **Clinical:** RTA pile-ups, earthquakes.

- **Examiner Tip:** Black = deceased/expectant, not "least urgent" misread.
- **Ref:** ATLS.

45. Sepsis & SIRS ★★★★★

- **Notes:** SIRS ≥ 2 : $T > 38$ or < 36 , $HR > 90$, $RR > 20$ (or $PaCO_2 < 32$), $WCC > 12$ or < 4 . Sepsis = SIRS + suspected infection. qSOFA: $RR \geq 22$, altered mentation, $SBP \leq 100$. Septic shock = refractory hypotension.
- **Rapid Rev:** SIRS needs 2 criteria. qSOFA = bedside quick screen.
- **SBA:** SIRS is diagnosed by at least: **2 of the four criteria**.
- **MCQ:** qSOFA includes: **$RR \geq 22$, altered mentation, $SBP \leq 100$** .
- **Explanation:** Sepsis bundles: cultures, ABx, fluids.
- **Trend:** High; definitions.
- **Clinical:** Post-op, UTI, pneumonia, wound sepsis.
- **Examiner Tip:** SIRS can occur WITHOUT infection (e.g., trauma, pancreatitis).
- **Ref:** Surviving Sepsis.

Next: `07-ORTHOPAEDICS-HIGH-YIELD.md` — topics #46–#100.

PART 3 — ORTHOPAEDICS HIGH-YIELD (Topics #46–#100)

Format: Notes · Rapid Rev · SBA/MCQ · Explanation · Trend · Clinical · Examiner Tip · Reference

Refs: Apley · Campbell · Rockwood & Green · AO Principles · Robbins · Miller's Review

46. Femoral Neck Fracture ★★★★★

- **Notes:** Intracapsular; Garden I–IV (I incomplete, II complete non-displaced, III partial displacement, IV full displacement). Displaced $\rightarrow$ disrupted MCFA retinacular vessels $\rightarrow$ AVN + non-union. Rx: < 60 ORIF (cannulated screws); > 65 arthroplasty (hemi/total).
- **Rapid Rev:** Garden $\uparrow$ = displacement $\uparrow$ = AVN $\uparrow$. Intracapsular = precarious blood.
- **SBA:** Most common complication of displaced intracapsular neck #: **avascular necrosis**.
- **MCQ:** Garden Type IV = **complete displacement**.
- **Explanation:** Capsular vessels torn $\rightarrow$ head ischaemia.
- **Trend:** Very high; Garden + AVN.
- **Clinical:** Elderly fall, osteoporosis.
- **Examiner Tip:** Intracapsular (Garden) = AVN risk; intertrochanteric = union good.
- **Ref:** Apley; Rockwood.

47. Intertrochanteric Fracture ★★★★★

- **Notes:** Extracapsular; rich blood supply $\rightarrow$ unites well. Evans classification (I stable, II unstable). Rx: Dynamic Hip Screw (DHS) / intramedullary nail (PFN). Mortality high (medical comorbidities).
- **Rapid Rev:** Extracapsular = good union; DHS for stable, IM nail for unstable.
- **SBA:** Treatment of choice for stable intertrochanteric #: **Dynamic Hip Screw (DHS)**.
- **MCQ:** Intertrochanteric # has: **good blood supply $\rightarrow$ unites well**.
- **Explanation:** Extracapsular vessels intact.

- **Trend:** Very high; fixation choice.
- **Clinical:** Elderly, osteoporosis.
- **Examiner Tip:** Unlike neck #, AVN is NOT a worry here.
- **Ref:** Campbell.

48. Femoral Shaft Fracture ★★★★★

- **Notes:** Winquist classification (comminution). Rx: closed intramedullary nailing (antegrade). Child: elastic stable IM nail (ESIN). Risk: haemorrhage (up to 1.5 L), fat embolism.
- **Rapid Rev:** IM nail = standard. Paediatric = ESIN/TEN.
- **SBA:** Definitive fixation of adult femoral shaft #: **intramedullary nailing**.
- **MCQ:** Winquist classification grades: **comminution of femoral shaft #**.
- **Explanation:** Nail = load-sharing, early mobilisation.
- **Trend:** Very high; fixation.
- **Clinical:** RTA, pathology #.
- **Examiner Tip:** Femoral # = major haemorrhage source.
- **Ref:** Rockwood.

49. Supracondylar Humerus Fracture ★★★★★

- **Notes:** Children (fall on outstretched hand). Gartland I (non-displaced), II (angulated, intact posterior cortex), III (fully displaced). Complications: anterior interosseous nerve (AIN, branch of median → cannot make "OK" sign), brachial artery injury, Volkmann ischaemia.
- **Rapid Rev:** Gartland III = displaced. AIN = "OK" sign lost.
- **SBA:** Nerve most often injured in supracondylar #: **anterior interosseous (median branch)**.
- **MCQ:** Volkmann ischaemic contracture follows: **untreated elevated compartment pressure (supracondylar #)**.
- **Explanation:** Swelling/cast compresses forearm compartments.
- **Trend:** Very high; nerve/artery.
- **Clinical:** 5–7 yr child.
- **Examiner Tip:** AIN = cannot flex IP thumb (no OK sign); not radial.
- **Ref:** Apley.

50. Colles' Fracture ★★★★★

- **Notes:** Distal radius metaphyseal # with dorsal displacement/shortening → "dinner-fork" deformity. Elderly osteoporotic women. Rx: closed reduction + plaster (below-elbow). Complication: malunion (dinner-fork), median nerve, CRPS, EPL rupture.
- **Rapid Rev:** "Dinner fork" = Colles (dorsal). Smith = volar (reverse).
- **SBA:** Dinner-fork deformity is characteristic of: **Colles' fracture**.
- **MCQ:** Colles' # is a fracture of: **distal radius (within 2–3 cm of wrist)**.
- **Explanation:** Dorsal angulation of distal fragment.
- **Trend:** Very high; image-based.
- **Clinical:** FOOSH, osteoporosis.
- **Examiner Tip:** Smith = volar (not dinner-fork).
- **Ref:** Apley.

51. Scaphoid Fracture ★★★★★

- **Notes:** FOOSH; tenderness in anatomical snuffbox. Waist most common; proximal pole = AVN risk (retrograde blood from dorsal carpal branch). Rx: thumb spica cast; non-union → scaphoid non-union advanced collapse (SNAC).
- **Rapid Rev:** Snuffbox tenderness = scaphoid. Proximal pole = AVN.
- **SBA:** Scaphoid # with AVN risk involves: **proximal pole**.
- **MCQ:** Best initial investigation if X-ray normal but snuffbox tender: **scaphoid views / MRI / repeat X-ray at 10–14 d**.
- **Explanation:** Early X-ray may miss; MRI sensitive.
- **Trend:** Very high; AVN + snuffbox.
- **Clinical:** Young adult FOOSH.
- **Examiner Tip:** Proximal pole (not waist) = highest AVN.
- **Ref:** Apley.

52. Tibial Plateau Fracture ★★★★★

- **Notes:** Schatzker I–VI (I lateral split, II depression, III posteromedial, IV medial/Tibial spine, V bicondylar, VI metaphyseal-diaphyseal dissociation). Association: ligament/meniscus injury, compartment syndrome. Rx: ORIF / external fixator.
- **Rapid Rev:** Schatzker = tibial plateau. VI = most severe.
- **SBA:** Schatzker classification is used for: **tibial plateau fractures**.
- **MCQ:** Tibial plateau # is commonly associated with: **meniscal/ligament injury & compartment syndrome**.
- **Explanation:** High-energy split/depression.
- **Trend:** High; classification.
- **Clinical:** Valgus load, elderly.
- **Examiner Tip:** Check knee ligaments in every plateau #.
- **Ref:** Schatzker; Rockwood.

53. Ankle Fracture ★★★★★

- **Notes:** Weber (A = below syndesmosis, B = at, C = above; C = worst, syndesmosis injured). Lauge-Hansen (supination-external rotation most common). Bimalleolar = medial+lat; trimalleolar + posterior. Rx: ORIF if displaced.
- **Rapid Rev:** Weber C = above syndesmosis = unstable.
- **SBA:** Most unstable ankle #: **Weber C (syndesmosis disrupted)**.
- **MCQ:** Lauge-Hansen most common pattern: **supination-external rotation**.
- **Explanation:** Higher Weber = more ligamentous injury.
- **Trend:** Very high; Weber classification.
- **Clinical:** Inversion/eversion injury.
- **Examiner Tip:** Weber B = at the syndesmosis (not below).
- **Ref:** Rockwood.

54. Calcaneal Fracture ★★★★★

- **Notes:** Fall from height; intra-articular (75%). Böhler's angle (normal 20–40°) ↓ on lateral X-ray. Sanders CT classification. Complications: subtalar arthritis, compartment syndrome.
- **Rapid Rev:** Böhler's angle <20° = #. Fall from height.
- **SBA:** Reduced Böhler's angle suggests: **calcaneal fracture**.
- **MCQ:** Mechanism of calcaneal #: **axial loading (fall from height)**.
- **Explanation:** Talus drives into calcaneus.
- **Trend:** High; Böhler's angle numerical.
- **Clinical:** Jump/fall.

- **Examiner Tip:** Always check spine (lumbar) with calcaneal # (axial load).
- **Ref:** Rockwood.

55. Hip Dislocation ★★★★★

- **Notes:** Posterior 90% (dashboard, flexed knee) → short, adducted, internally rotated, flexed. Sciatic nerve (common peroneal) injured. Anterior → externally rotated, abducted. Reduce urgently (<6–12 h) to avoid AVN.
- **Rapid Rev:** Posterior = short + IR + adducted; sciatic nerve.
- **SBA:** Most common hip dislocation: **posterior (90%)**.
- **MCQ:** Nerve injured in posterior hip dislocation: **sciatic (common peroneal branch)**.
- **Explanation:** Femoral head exits posteriorly → stretches sciatic.
- **Trend:** Very high; position + nerve.
- **Clinical:** RTA.
- **Examiner Tip:** Posterior = internal rotation; anterior = external rotation.
- **Ref:** Apley.

56. Pelvic Fracture ★★★★★

- **Notes:** Tile A (stable), B (partial stable, open-book/APC), C (unstable, vertical shear/LC). Major haemorrhage (retroperitoneal, venous). Pelvic binder; embolisation; APR if open.
- **Rapid Rev:** Tile C = unstable/vertical shear. Binder first.
- **SBA:** Unstable vertical-shear pelvic #: **Tile C**.
- **MCQ:** Initial stabilisation of unstable pelvic #: **pelvic binder / external fixator**.
- **Explanation:** Binder reduces volume, tamponades bleeding.
- **Trend:** High; Tile + haemorrhage.
- **Clinical:** High-energy RTA.
- **Examiner Tip:** Open-book (APC) = external rotation; bind immediately.
- **Ref:** Rockwood; ATLS.

57. Open Fracture ★★★★★

- **Notes:** Gustilo-Anderson I (<1 cm, clean), II (1–10 cm), IIIA (severe soft tissue but coverable), IIIB (needs flap), IIIC (arterial injury). Rx: urgent debridement <6–8 h, ABx (cefazolin ± aminoglycoside ± penicillin), tetanus, stabilise, soft-tissue coverage.
- **Rapid Rev:** Gustilo III = high energy. IIIC = vessel.
- **SBA:** Open # with arterial injury: **Gustilo IIIC**.
- **MCQ:** Antibiotics for open # should be given: **as early as possible (within 1 h), pre-debridement**.
- **Explanation:** Debridement + early ABx ↓ infection.
- **Trend:** Very high; Gustilo + management.
- **Clinical:** RTA, farming injuries.
- **Examiner Tip:** Debridement timing is the single biggest infection determinant.
- **Ref:** AO Principles; Gustilo.

58. Paediatric Fractures ★★★★★

- **Notes:** Greenstick (incomplete, bent), torus/buckle (impaction), plastic deformity. Salter-Harris (physeal): I transverse through physis, II (most common, metaphysis + physis), III (physis + epiphysis, intra-articular), IV (through all 3), V (crush physis). Complications: growth arrest (III/IV/V).

- **Rapid Rev:** SALTR mnemonic: I Straight, II Above, III Lower (epi), IV Through, V cRush.
- **SBA:** Most common Salter-Harris #: **Type II**.
- **MCQ:** Type V Salter-Harris is: **compression/crush injury of physis (growth arrest)**.
- **Explanation:** Physeal plate damage → limb length discrepancy.
- **Trend:** Very high; Salter-Harris.
- **Clinical:** Child fractures.
- **Examiner Tip:** Type II = "Above" metaphysis; most common.
- **Ref:** Apley.

59. Fat Embolism ★★★★★

- **Notes:** After long-bone # (femur/tibia) / pelvic. Triad: respiratory distress, petechial rash (conjunctiva/upper body), CNS confusion. Supportive; prophylaxis = early fixation.
- **Rapid Rev:** Petechiae + dyspnoea + long-bone # = fat embolism.
- **SBA:** Fat embolism typically follows: **long-bone (femur/tibia) fractures**.
- **MCQ:** Petechiae in fat embolism appear over: **conjunctivae & upper body**.
- **Explanation:** Fat globules → pulmonary + systemic emboli.
- **Trend:** Very high; triad.
- **Clinical:** Multiple #, IM nailing.
- **Examiner Tip:** Petechiae = diagnostic clue (not just hypoxia).
- **Ref:** Apley.

60. DVT/PE in Ortho ★★★★★

- **Notes:** Highest risk after hip/knee arthroplasty, pelvis, femur #. Prophylaxis LMWH ± IPC. Symptom: calf pain/swelling; PE = hypoxia/chest pain. Wells score.
- **Rapid Rev:** Chemoprophylaxis until mobilised (esp. THR/TKR).
- **SBA:** Highest VTE risk ortho procedure: **hip/knee replacement & pelvis #**.
- **MCQ:** Standard VTE prophylaxis post-THR: **LMWH + early mobilisation + stockings**.
- **Explanation:** Immobility + hypercoagulability.
- **Trend:** High; overlaps #38.
- **Clinical:** Post-op.
- **Examiner Tip:** Aspirin alone insufficient for high-risk.
- **Ref:** AAOS.

61. DDH ★★★★★

- **Notes:** Developmental dysplasia of hip. Barlow (dislocatable), Ortolani (reduces click) <3 mo. Graf ultrasound <6 mo; X-ray after (acetabular index, Shenton line). Rx: Pavlik harness (infants), closed/open reduction + spica.
- **Rapid Rev:** Ortolani = CLICK reducible; Barlow = dislocatable. USG <6 mo.
- **SBA:** Positive Ortolani sign means: **dislocated hip can be reduced (click)**.
- **MCQ:** Investigation of choice for DDH <6 months: **ultrasound (Graf)**.
- **Explanation:** Cartilage not ossified → USG better than X-ray.
- **Trend:** Very high; tests + imaging.
- **Clinical:** Female, breech, firstborn.
- **Examiner Tip:** Ortolani = reducible click; Barlow = dislocatable (different!).
- **Ref:** Apley.

62. CTEV (Clubfoot) ★★★★★

- **Notes:** Congenital talipes equinovarus: Cavus, Adductus, Varus, Equinus. Pirani / Dimeglio score. Ponseti method = serial casting + tenotomy of Achilles; boots-and-bars maintenance.
- **Rapid Rev:** "CAVE": Cavus, Adductus, Varus, Equinus. Ponseti = gold standard.
- **SBA:** Treatment of choice for idiopathic clubfoot: **Ponseti serial casting**.
- **MCQ:** Ponseti method includes: **serial casts + percutaneous Achilles tenotomy**.
- **Explanation:** Gradual correction; tenotomy for equinus.
- **Trend:** Very high; Ponseti.
- **Clinical:** Newborn.
- **Examiner Tip:** Maintenance brace prevents relapse (most recurrences = non-compliance).
- **Ref:** Apley.

63. Cerebral Palsy ★★★★★

- **Notes:** Non-progressive motor disorder (perinatal brain injury). Spastic (most), dyskinetic, ataxic. GMFCS levels I–V. Management: botulinum toxin, serial casting, selective dorsal rhizotomy, multilevel surgery.
- **Rapid Rev:** GMFCS I (walks) → V (wheelchair). Spastic = commonest.
- **SBA:** Most common CP type: **spastic (diplegic/hemiplegic/quadruplegic)**.
- **MCQ:** Botulinum toxin in CP is used for: **spasticity (dynamic contracture)**.
- **Explanation:** Chemodenervation reduces tone.
- **Trend:** High; GMFCS.
- **Clinical:** Paediatric ortho.
- **Examiner Tip:** CP is NON-progressive (lesion static; manifestations change).
- **Ref:** Campbell.

64. Physeal Growth ★★★

- **Notes:** Epiphyseal plate = growth; proximal tibia/distal femur contribute most lower-limb length. Injury → partial arrest (angular) or complete (shortening). Guided growth (8-plate) for angular.
- **Rapid Rev:** Most growth: distal femur + proximal tibia.
- **SBA:** Greatest lower-limb growth from: **distal femoral & proximal tibial physes**.
- **MCQ:** Angular deformity after physeal bar treated by: **guided growth / bar resection**.
- **Explanation:** Asymmetric growth → deformity.
- **Trend:** Moderate; growth plates.
- **Clinical:** Child limb discrepancy.
- **Examiner Tip:** Distal femur physis = ~40% femoral growth.
- **Ref:** Apley.

65. SCFE ★★★★★

- **Notes:** Slipped capital femoral epiphysis (adolescent obese male). Klein's line (should intersect femoral head on AP) fails → slip. Rx: pins-in-situ (percutaneous); avoid manipulative reduction (AVN risk).
- **Rapid Rev:** Klein's line through superior neck should cross head.
- **SBA:** Investigation showing SCFE: **AP pelvis with Klein's line not crossing femoral head**.
- **MCQ:** Treatment of stable SCFE: **in-situ pinning (screw)**.
- **Explanation:** Fix slip, do not reduce (AVN/CHD risk).
- **Trend:** Very high; Klein + management.

- **Clinical:** Obese adolescent, limp.
- **Examiner Tip:** Do NOT reduce the slip (causes AVN).
- **Ref:** Apley.

66. Perthes Disease ★★★★★

- **Notes:** Idiopathic AVN of femoral head (4–8 yr, boys). Waldenström phases (necrosis→fragmentation→reossification→healed). Herring/Lateral-pillar classification (A/B/C). Containment (abduction brace/osteotomy) if >6 yr or C.
- **Rapid Rev:** Age >6 + lateral pillar C = poor prognosis.
- **SBA:** Lateral-pillar (Herring) C implies: **poor prognosis**.
- **MCQ:** Perthes occurs typically in: **boys 4–8 years**.
- **Explanation:** AVN remodels; older age = worse.
- **Trend:** Very high; classification.
- **Clinical:** Limp, restricted abduction/IR.
- **Examiner Tip:** Perthes = AVN in CHILD (not adult steroid AVN).
- **Ref:** Apley.

67. Rickets / Osteomalacia ★★★★★

- **Notes:** Vit-D deficiency → defective mineralisation. Children: wide epiphyses, rachitic rosary, craniotables, bowed legs, Harrison's sulcus. Adults: bone pain, Looser's zones (pseudofractures), proximal myopathy.
- **Rapid Rev:** Rickets = child (open plates); osteomalacia = adult.
- **SBA:** Rachitic rosary is seen in: **rickets (vitamin-D deficiency)**.
- **MCQ:** Looser's zones are characteristic of: **osteomalacia**.
- **Explanation:** Unmineralised osteoid → pseudofractures.
- **Trend:** Very high; BD endemic.
- **Clinical:** Nutritional, malabsorption, renal.
- **Examiner Tip:** Rickets (children) vs osteomalacia (adults) — same deficiency.
- **Ref:** Robbins.

68. Osteogenesis Imperfecta ★★★★★

- **Notes:** Type I collagen defect → brittle bones, blue sclerae, hearing loss, dentinogenesis imperfecta, wormian bones. Rx: bisphosphonates (↓ fracture), physical therapy, rodding.
- **Rapid Rev:** "Blue sclerae + multiple #" = OI.
- **SBA:** Blue sclerae are characteristic of: **osteogenesis imperfecta**.
- **MCQ:** OI is due to defect in: **type I collagen**.
- **Explanation:** Faulty collagen → fragile bone.
- **Trend:** High; type + sign.
- **Clinical:** Recurrent childhood fractures.
- **Examiner Tip:** OI = Type I (not II).
- **Ref:** Robbins.

69. Achondroplasia & Dysplasias ★★★★★

- **Notes:** Most common dwarfism; FGFR3 mutation → inhibited endochondral ossification → short limbs, normal trunk, macrocephaly, rhizomelic shortening. Foramen magnum/spinal stenosis risk.

- **Rapid Rev:** "Short limbs, normal trunk" = achondroplasia.
- **SBA:** Achondroplasia causes: **defective endochondral ossification (FGFR3)**.
- **MCQ:** In achondroplasia, trunk is: **normal / relatively long (limbs short)**.
- **Explanation:** Membranous (skull/trunk) spared; endochondral (long bones) impaired.
- **Trend:** High; features.
- **Clinical:** Dwarfism.
- **Examiner Tip:** Rhizomelic = proximal segments shorter.
- **Ref:** Robbins.

70. Congenital Pseudoarthrosis Tibia ★★★

- **Notes:** Rare; anterolateral bow + pseudoarthrosis (often NF1). Rx: Ilizarov, intramedullary rod, bone graft; high non-union.
- **Rapid Rev:** Tibial anterolateral bow + pseudoarthrosis.
- **SBA:** Congenital pseudoarthrosis is most common in: **tibia**.
- **MCQ:** Associated condition: **neurofibromatosis type 1**.
- **Explanation:** Dysplastic anterior tibia → fracture → non-union.
- **Trend:** Moderate; rare.
- **Clinical:** Child.
- **Examiner Tip:** Think NF1 when you see it.
- **Ref:** Apley.

71. Cervical Spine ★★★★★

- **Notes:** Cord syndromes: central (central canal → upper > lower limb, hyperextension in spondylosis), Brown-Séquard (hemisection → ipsilateral motor+position, contralateral pain/temp). Atlanto-axial (odontoid) # in elderly falls.
- **Rapid Rev:** Central = arms worse; Brown-Séquard = same-side motor.
- **SBA:** Ipsilateral weakness + contralateral pain loss = **Brown-Séquard syndrome**.
- **MCQ:** Hyperextension in cervical spondylosis → **central cord syndrome**.
- **Explanation:** Central grey matter affected → upper limb > lower.
- **Trend:** Very high; syndromes.
- **Clinical:** Elderly fall, trauma.
- **Examiner Tip:** Central cord = upper limbs affected more.
- **Ref:** Apley.

72. Lumbar Disc Prolapse ★★★★★

- **Notes:** L4–L5 (L5 root) & L5–S1 (S1 root) most common. Sciatica; straight-leg raise (SLR) +ve 30–60°; crossed SLR specific. Rx: conservative; surgery if cauda equina/deficit.
- **Rapid Rev:** SLR +ve = disc. L5 = foot drop/big toe; S1 = plantarflexion/ankle jerk.
- **SBA:** Positive straight-leg raise indicates: **lumbar disc prolapse**.
- **MCQ:** Most common level of disc prolapse: **L4–L5**.
- **Explanation:** Posterolateral herniation compresses traversing root.
- **Trend:** Very high; SLR.
- **Clinical:** Sciatica.
- **Examiner Tip:** Crossed SLR (opposite leg) = more specific.

- **Ref:** Apley.

73. Spondylolisthesis ★★★★★

- **Notes:** Anterior slip of vertebra (isthmic = pars defect L5/S1; degenerative). Meyerding I–IV ($\leq 25\%$ to $>75\%$). Slipped $>50\%$ (III/IV) → surgery.
- **Rapid Rev:** Isthmic = pars interarticularis defect.
- **SBA:** Isthmic spondylolisthesis is due to: **pars interarticularis defect**.
- **MCQ:** Meyerding grade III = slip of: **50–75%**.
- **Explanation:** Slip graded by % of vertebral width.
- **Trend:** High; classification.
- **Clinical:** Spondylolysis (pars #) in athletes.
- **Examiner Tip:** Pars defect = isthmic (not degenerative).
- **Ref:** Apley.

74. Scoliosis ★★★★★

- **Notes:** Lateral curvature + rotation. Idiopathic (adolescent, girls), congenital, neuromuscular. Cobb angle ($>10^\circ$ diagnostic). Bracing (Milwaukee/TLSO) for $25\text{--}45^\circ$ growing; surgery $>45\text{--}50^\circ$.
- **Rapid Rev:** Cobb angle = standard measure. Brace $25\text{--}45^\circ$, surgery $>45^\circ$.
- **SBA:** Scoliosis is measured by: **Cobb angle**.
- **MCQ:** Indication for bracing in adolescent idiopathic scoliosis: **$25\text{--}45^\circ$ with growth remaining**.
- **Explanation:** Halts progression until skeletal maturity.
- **Trend:** Very high; Cobb + brace.
- **Clinical:** Screening adolescent girls.
- **Examiner Tip:** Cobb $>10^\circ$ = scoliosis; $>45^\circ$ = surgery.
- **Ref:** Apley.

75. Kyphosis & Scheuermann's ★★★★★

- **Notes:** Normal thoracic kyphosis $20\text{--}40^\circ$; $>50^\circ$ pathological. Scheuermann's = anterior wedging $\geq 5^\circ$ of ≥ 3 vertebrae (juvenile); postural vs structural.
- **Rapid Rev:** Scheuermann = wedging of 3+ vertebrae.
- **SBA:** Scheuermann's kyphosis shows: **anterior wedging of ≥ 3 vertebrae $\geq 5^\circ$** .
- **MCQ:** Structural kyphosis $>50^\circ$ suggests: **pathological (Scheuermann's / TB)**.
- **Explanation:** Vertebral wedging → rigid kyphosis.
- **Trend:** Moderate; definition.
- **Clinical:** Adolescent.
- **Examiner Tip:** Postural kyphosis corrects on extension; Scheuermann's does not.
- **Ref:** Apley.

76. Spinal TB (Pott's) ★★★★★

- **Notes:** Mycobacterium tuberculosis; paradiscal spread from vertebral body → disc destruction + vertebral collapse → angular (gibbus) kyphosis; cold abscess (psoas/paravertebral); cord/paraplegia. RX: ATT 12–18 mo; surgery if neurologic/instability.
- **Rapid Rev:** "Cold abscess + angular kyphosis" = Pott's. ATT first.
- **SBA:** Angular kyphosis with cold abscess suggests: **spinal tuberculosis (Pott's)**.

- **MCQ:** Spread in spinal TB is: **paradiscal (from body to disc)**.
- **Explanation:** Disc destroyed → collapse → gibbus.
- **Trend:** Very high; BD endemic.
- **Clinical:** Dorsal spine, paraplegia.
- **Examiner Tip:** Pott's = COLD abscess (no warmth/redness).
- **Ref:** Apley; Robbins.

77. Spinal Stenosis ★★★★★

- **Notes:** Narrowing of spinal canal → neurogenic claudication (pain on standing/walking, relieved by sitting/forward flexion). Lumbar common. Rx: decompressive laminectomy.
- **Rapid Rev:** Neurogenic claudication = spinal stenosis; vascular = calf, relieved by rest not posture.
- **SBA:** Pain on standing relieved by flexion = **spinal stenosis (neurogenic claudication)**.
- **MCQ:** Spinal stenosis is best confirmed by: **MRI/CT**.
- **Explanation:** Position-dependent symptoms = hallmark.
- **Trend:** High; claudication differentiation.
- **Clinical:** Elderly.
- **Examiner Tip:** Vascular claudication = calf, rest relieves; neurogenic = posture relieves.
- **Ref:** Apley.

78. Cauda Equina Syndrome ★★★★★

- **Notes:** Emergency: saddle anaesthesia, bladder/bowel incontinence, bilateral radiculopathy, weak S1. Causes: massive disc, tumour, trauma, abscess. Rx: urgent decompression (<48 h).
- **Rapid Rev:** Saddle anaesthesia + urinary retention = RED FLAG.
- **SBA:** Cauda equina syndrome includes: **saddle anaesthesia + urinary retention**.
- **MCQ:** Management of cauda equina: **urgent surgical decompression**.
- **Explanation:** Permanent incontinence if delayed.
- **Trend:** Very high; emergency.
- **Clinical:** Large central L-disc, tumour.
- **Examiner Tip:** Urinary retention = surgical emergency (not conservative).
- **Ref:** Apley.

79. Osteoarthritis ★★★★★

- **Notes:** Degenerative; cartilage loss, osteophytes, subchondral sclerosis/cysts, joint-space narrowing (JSN). Knee > hip > hand. Kellgren-Lawrence grading. Rx: weight loss, physio, analgesia, joint replacement.
- **Rapid Rev:** JSN + osteophytes = OA. Heberden (DIP)/Bouchard (PIP) nodes.
- **SBA:** Radiologic feature of OA: **joint-space narrowing + osteophytes**.
- **MCQ:** Kellgren-Lawrence classifies: **severity of osteoarthritis**.
- **Explanation:** Cartilage wear → bone-on-bone.
- **Trend:** Very high; X-ray signs.
- **Clinical:** Elderly, obesity.
- **Examiner Tip:** OA = JSN (not osteoporosis). MCP spared (unlike RA).
- **Ref:** Apley.

80. Rheumatoid Arthritis ★★★★★

- **Notes:** Autoimmune, symmetrical polyarthritis; MCP/PIP, wrist; swan-neck (PIP ext, DIP flex), boutonnière (PIP flex, DIP ext); erosions, ulnar deviation. Cervical subluxation (atlanto-axial) = airway risk. DMARDs (MTX), biologics.
- **Rapid Rev:** RA = MCP/PIP (NOT DIP); Swan-neck vs Boutonnière.
- **SBA:** Cervical subluxation is a complication of: **rheumatoid arthritis**.
- **MCQ:** Swan-neck deformity = **PIP hyperextension + DIP flexion**.
- **Explanation:** Synovitis erodes joints → deformities.
- **Trend:** Very high; deformities + C-spine.
- **Clinical:** Hands, feet.
- **Examiner Tip:** RA spares DIP (PsA affects DIP).
- **Ref:** Apley.

81. Septic Arthritis ★★★★★

- **Notes:** Hot, swollen, restricted joint; neonate/elderly, immuno-compromised. *S. aureus* #1; gonococcus in young sexually active; *H. influenzae* in unvaccinated children. Dx: aspirate (Gram, culture) BEFORE ABx; urgent drainage + ABx.
- **Rapid Rev:** "Hot joint → aspirate first, then ABx." Knee common.
- **SBA:** Investigation of choice for septic arthritis: **joint aspiration + culture**.
- **MCQ:** Most common organism in septic arthritis: ***S. aureus***.
- **Explanation:** Bacteria seed synovium → purulent arthritis.
- **Trend:** Very high; organism + aspiration.
- **Clinical:** Knee, hip (hip = urgent).
- **Examiner Tip:** Aspirate BEFORE starting antibiotics.
- **Ref:** Apley; Robbins.

82. Gout & Pseudogout ★★★★★

- **Notes:** Gout = monosodium urate, negatively birefringent needles, 1st MTP (podagra), tophi. Pseudogout (CPPD) = calcium pyrophosphate, positively birefringent rhomboids, knee, chondrocalcinosis. Acute: NSAIDs/colchicine/steroids; prophylaxis: allopurinol/febuxostat.
- **Rapid Rev:** Gout = NEGATIVE needles; Pseudogout = POSITIVE rhomboids.
- **SBA:** Negatively birefringent crystals = **gout**.
- **MCQ:** Pseudogout crystals are: **calcium pyrophosphate (positive birefringence)**.
- **Explanation:** Crystal type differentiates.
- **Trend:** Very high; crystal sign.
- **Clinical:** Podagra, renal stones.
- **Examiner Tip:** Pseudogout = knees + chondrocalcinosis (not 1st MTP).
- **Ref:** Robbins.

83. Ankylosing Spondylitis ★★★★★

- **Notes:** Seronegative spondyloarthropathy; HLA-B27; sacroiliitis → bamboo spine; enthesitis; uveitis; reduced chest expansion. Schober test <4 cm (normal ≥5). Rx: NSAIDs, physiotherapy, TNF inhibitors.
- **Rapid Rev:** Bamboo spine + HLA-B27. Schober <4 cm.
- **SBA:** Bamboo spine is characteristic of: **ankylosing spondylitis**.
- **MCQ:** Schober test assesses: **lumbar flexion (AS limited)**.
- **Explanation:** Spinal fusion → stiffness.
- **Trend:** Very high; HLA-B27 + Schober.

- **Clinical:** Young male, back stiffness.
- **Examiner Tip:** Schober <4 cm = limited flexion (not >4).
- **Ref:** Apley.

84. Avascular Necrosis ★★★★★

- **Notes:** Ischaemic necrosis; femoral head (steroids, alcohol, trauma, SLE), scaphoid proximal pole, talus, medial femoral condyle. MRI earliest. Rx: core decompression early, arthroplasty late.
- **Rapid Rev:** "ASEPTIC": Alcohol, Steroids, Trauma, Idiopathic, Caisson, Lupus, Sickle.
- **SBA:** Gold-standard investigation for early AVN: **MRI**.
- **MCQ:** Most common cause of femoral-head AVN: **steroids / fracture neck femur**.
- **Explanation:** X-ray normal early; MRI shows oedema.
- **Trend:** Very high; MRI + causes.
- **Clinical:** Steroid use, trauma.
- **Examiner Tip:** MRI, not X-ray, for early AVN.
- **Ref:** Apley.

85. Charcot Joint ★★★★★

- **Notes:** Neuropathic joint destruction from lost sensation (diabetes, syphilis, spinal cord injury). Pain-disproportionate to X-ray destruction; "bag of bones". Rx: offloading, braces; surgery late.
- **Rapid Rev:** "Looks terrible, hurts little" = Charcot.
- **SBA:** Charcot joint is due to: **neuropathy (loss of sensation)**.
- **MCQ:** Most common cause of Charcot joint today: **diabetes mellitus**.
- **Explanation:** Repeated unnoticed trauma → destruction.
- **Trend:** High; diabetes link.
- **Clinical:** Diabetic foot.
- **Examiner Tip:** Charcot = neuropathic (not septic) though can superinfect.
- **Ref:** Apley.

86. Hemophilic Arthropathy ★★★★★

- **Notes:** Recurrent haemarthrosis → synovitis, cartilage loss, arthropathy (knee/elbow/ankle). Prophylactic factor VIII/IX. Rx: factor replacement, aspiration (caution), synovectomy.
- **Rapid Rev:** Haemarthrosis → "target joint."
- **SBA:** Hemophilic arthropathy results from: **recurrent intra-articular haemorrhage**.
- **MCQ:** Target joints in haemophilia: **knee, elbow, ankle**.
- **Explanation:** Iron deposition → synovitis → damage.
- **Trend:** High; mechanism.
- **Clinical:** Haemophilia A/B.
- **Examiner Tip:** Prophylactic factor prevents arthropathy.
- **Ref:** Apley.

87. ACL / PCL ★★★★★

- **Notes:** ACL: ant drawer, Lachman (most sensitive), pivot shift; non-contact valgus+rotation. PCL: post drawer; dashboard. Rx: reconstruction (hamstring/patellar graft).
- **Rapid Rev:** Lachman = best test. Pivot shift = rotatory.

- **SBA:** Most sensitive test for ACL tear: **Lachman test**.
- **MCQ:** Mechanism of ACL injury: **non-contact pivoting / valgus rotation**.
- **Explanation:** ACL resists anterior tibial translation.
- **Trend:** Very high; Lachman.
- **Clinical:** Sports.
- **Examiner Tip:** Lachman > anterior drawer (more sensitive).
- **Ref:** Apley.

88. Meniscal Injury ★★★★★

- **Notes:** Medial (>lateral) due to MCL attachment. McMurray (click), Apley (compression/rotation), Thessaly. Bucket-handle = locked knee. Rx: arthroscopic repair (peripheral) / partial meniscectomy.
- **Rapid Rev:** Medial meniscus torn more; bucket-handle = lock.
- **SBA:** Bucket-handle meniscal tear causes: **locked knee**.
- **MCQ:** McMurray test assesses: **meniscal tear**.
- **Explanation:** Torn fragment clicks on rotation.
- **Trend:** Very high; tests.
- **Clinical:** Twisting injury.
- **Examiner Tip:** Medial meniscus = fixed = more injured.
- **Ref:** Apley.

89. Rotator Cuff ★★★★★

- **Notes:** Supraspinatus (abduction 0–15°, impingement), infraspinatus/teres minor (ext rot), subscapularis (int rot). Supraspinatus most torn. Drop-arm test; painful arc 60–120°. Rx: physio, subacromial decompression, repair.
- **Rapid Rev:** "SITS". Supraspinatus = most commonly torn/impinged.
- **SBA:** Most commonly torn rotator cuff tendon: **supraspinatus**.
- **MCQ:** Painful arc (60–120°) suggests: **subacromial impingement / supraspinatus tendinopathy**.
- **Explanation:** Tendon compressed under acromion in mid-range.
- **Trend:** Very high; cuff anatomy.
- **Clinical:** Overhead activity, elderly.
- **Examiner Tip:** Drop-arm = supraspinatus/large cuff tear.
- **Ref:** Apley.

90. Tendinopathy ★★★★★

- **Notes:** Tennis elbow = lateral epicondylitis (extensor carpi radialis brevis); golfer's = medial. Achilles tendinopathy/rupture (Thompson test -ve squeeze). Rx: eccentric exercise, rest, NSAID, surgery.
- **Rapid Rev:** Tennis = lateral; golfer = medial. Thompson -ve = Achilles rupture.
- **SBA:** Tennis elbow involves: **lateral epicondyle (ECRB)**.
- **MCQ:** Positive Thompson test (no plantarflex on calf squeeze) = **Achilles tendon rupture**.
- **Explanation:** Loss of continuity → no reflex plantarflex.
- **Trend:** High; epicondyle + Thompson.
- **Clinical:** Overuse.
- **Examiner Tip:** Thompson -ve (absent movement) = rupture.
- **Ref:** Apley.

91. Osteosarcoma ★★★★★

- **Notes:** Most common primary malignant bone tumour in children/young. Metaphysis (distal femur, proximal tibia, proximal humerus). Sunburst periosteal reaction, Codman triangle. Lung metastases. Rx: neoadjuvant chemo + wide resection.
- **Rapid Rev:** Metaphysis + sunburst + Codman = osteosarcoma.
- **SBA:** Sunburst appearance on X-ray = **osteosarcoma**.
- **MCQ:** Most common primary malignant bone tumour in children: **osteosarcoma**.
- **Explanation:** Aggressive osteoid-producing tumour.
- **Trend:** Very high; imaging sign.
- **Clinical:** Adolescent knee pain/swelling.
- **Examiner Tip:** Osteosarcoma = METAPHYSIS (not diaphysis).
- **Ref:** Robbins; Campbell.

92. Ewing's Sarcoma ★★★★★

- **Notes:** Diaphysis (long bones, pelvis); onion-skin periosteal reaction; t(11;22) EWS-FLI1; <20 yr; fever/ESR↑ (mimics infection). Rx: chemo + radiation/surgery.
- **Rapid Rev:** Diaphysis + onion-skin + t(11;22) = Ewing's.
- **SBA:** Onion-skin periosteal reaction = **Ewing's sarcoma**.
- **MCQ:** Chromosomal translocation in Ewing's: **t(11;22)**.
- **Explanation:** Small round blue cells.
- **Trend:** Very high; sign + genetics.
- **Clinical:** Child/teen, fever.
- **Examiner Tip:** Ewing = DIAPHYSIS (vs osteosarcoma metaphysis).
- **Ref:** Robbins.

93. Osteomyelitis (Paediatric) ★★★★★

- **Notes:** Acute haematogenous; metaphysis (femur/distal femur); *S. aureus*. Neonates: GBS/*E. coli*. Sick: *Salmonella*. Subacute: Brodie's abscess. Chronic: sequestrum + involucrum.
- **Rapid Rev:** Kids = metaphysis; sequestrum = dead bone.
- **SBA:** Most common site of paediatric haematogenous osteomyelitis: **metaphysis (distal femur)**.
- **MCQ:** Sequestrum is: **necrotic dead bone within infected focus**.
- **Explanation:** Ischaemic bone dies → sequestrum.
- **Trend:** Very high; site + terms.
- **Clinical:** Fever, localised pain, limp.
- **Examiner Tip:** Involucrum = new living shell (not dead).
- **Ref:** Robbins.

94. Bone Metastasis ★★★★★

- **Notes:** Most common malignant bone tumour overall. Prostate = osteoblastic (sclerotic). Breast/lung/thyroid/kidney = lytic. Rx: bisphosphonate, radio, stabilisation.
- **Rapid Rev:** Prostate = blastic; breast/lung = lytic.
- **SBA:** Sclerotic (blastic) metastasis suggests: **prostate carcinoma**.
- **MCQ:** Most common bone metastases are from: **breast, prostate, lung, kidney, thyroid**.
- **Explanation:** Blastic vs lytic pattern hints primary.

- **Trend:** Very high; prostate/breast.
- **Clinical:** Back pain, pathological #.
- **Examiner Tip:** Prostate = blastic (not lytic).
- **Ref:** Robbins.

95. Giant-Cell Tumour ★★★★★

- **Notes:** Epiphysis (around knee — distal femur/proximal tibia), 20–40 yr. Soap-bubble / expansile lytic; Campanacci grading. "Osteoclastoma." Rx: curettage + cement/graft; denosumab; malignant transformation rare.
- **Rapid Rev:** Epiphysis + soap-bubble = GCT.
- **SBA:** Soap-bubble appearance = **giant-cell tumour**.
- **MCQ:** Giant-cell tumour typically occurs in: **epiphysis (around knee)**.
- **Explanation:** Locally aggressive, rarely malignant.
- **Trend:** Very high; location + sign.
- **Clinical:** Young adult knee.
- **Examiner Tip:** GCT = EPIPHYSIS (not metaphysis like osteosarcoma).
- **Ref:** Robbins.

96. Soft-Tissue Sarcoma ★★★★★

- **Notes:** Deep, >5 cm, painful, fixed; staging size/depth/grade + metastasis (lung). Rx: incisional biopsy then wide (compartment) excision ± radiotherapy; limb-salvage.
- **Rapid Rev:** "Deep + >5 cm" → sarcoma workup.
- **SBA:** Management principle for suspected soft-tissue sarcoma: **planned incisional biopsy then wide excision**.
- **MCQ:** Most common site of metastasis: **lung**.
- **Explanation:** Bad biopsy worsens outcome.
- **Trend:** High; biopsy principle.
- **Clinical:** Extremity mass.
- **Examiner Tip:** Plan biopsy with oncologic surgeon.
- **Ref:** Schwartz; Campbell.

97. Polytrauma (Ortho) ★★★★★

- **Notes:** Multiple injuries; ATLS; damage-control ortho (external fixator temporising for femur/pelvis); prioritise life > limb; early total care vs damage control.
- **Rapid Rev:** Life before limb. Temporary fixator then definitive.
- **SBA:** In polytrauma with femur #, initial stabilisation: **external fixator (damage control)**.
- **MCQ:** Damage-control ortho aims to: **temporise fixation to allow resuscitation**.
- **Explanation:** Avoids prolonged anaesthesia in unstable patient.
- **Trend:** High; trauma.
- **Clinical:** RTA.
- **Examiner Tip:** Convert to IM nail once stable.
- **Ref:** ATLS; Rockwood.

98. Osteoporosis ★★★★★

- **Notes:** Low bone mass; DEXA T-score ≤ -2.5 . Post-menopausal (type I) & senile (type II). Rx: bisphosphonate, Ca^{2+} /vit D, fall prevention. Atypical femur # (bisphosphonate). Vertebral/wrist/hip #.

- **Rapid Rev:** T-score ≤ -2.5 = osteoporosis. T -1 to -2.5 = osteopenia.
- **SBA:** Diagnosis of osteoporosis: **DEXA T-score ≤ -2.5 .**
- **MCQ:** Bisphosphonates are first-line for: **osteoporosis.**
- **Explanation:** $\downarrow$ resorption $\rightarrow \uparrow$ density.
- **Trend:** Very high; DEXA.
- **Clinical:** Elderly women.
- **Examiner Tip:** Osteopenia = T -1 to -2.5 (not osteoporosis).
- **Ref:** Apley.

99. Imaging in Ortho ★★★★★

- **Notes:** X-ray: 2 views (AP/lat) always; stress/oblique as needed. CT: complex #, pelvis, spine. MRI: soft tissue, AVN, disc, tumour, infection. Bone scan: metastases, stress #, occult. USG: DDH <6 mo, soft tissue, effusion.
- **Rapid Rev:** X-ray 1st; MRI soft tissue/AVN; bone scan metastases.
- **SBA:** Investigation of choice for AVN: **MRI.**
- **MCQ:** DDH <6 months: **ultrasound.**
- **Explanation:** Modality matched to tissue.
- **Trend:** Very high; "investigation of choice" stems.
- **Clinical:** All ortho.
- **Examiner Tip:** Bone scan sensitive but not specific.
- **Ref:** Apley.

100. Principles of Fracture Fixation ★★★★★

- **Notes:** AO/ASIF principles: anatomical reduction, stable fixation, preservation of blood supply (biological), early motion. Absolute stability (compression plate) vs relative stability (IM nail/ex-fix). ORIF vs MIPO/BO (biological osteosynthesis). Tension band (patella, olecranon).
- **Rapid Rev:** AO = reduce, fix, preserve biology, move early.
- **SBA:** AO principle emphasising biology: **preserve blood supply (minimally invasive / biological fixation).**
- **MCQ:** Tension-band principle used for: **patella & olecranon fractures.**
- **Explanation:** Converts tension $\rightarrow$ compression at fracture.
- **Trend:** Very high; principles.
- **Clinical:** Every #.
- **Examiner Tip:** Absolute stability = compression; relative = IM nail.
- **Ref:** AO Principles.

Next: `08-SBA-MCQ-BANK.md` (200+ Q) and `09-RAPID-REVISION-EXAM-NIGHT.md`.

PART 4 — SBA / MCQ BANK (High-Yield & Repeating Questions)

Examiner-style single-best-answer stems with answers + "Why". Use for daily 20-Q drill.

Section A: Basic Sciences · B: Faculty Surgery · C: Fractures/Trauma · D: Paediatric · E: Spine · F: Joints/Arthritis · G: Sports/Soft Tissue · H: Tumours/Infection · I: Image-Based

A. Basic Sciences (1–30)

Q1. Waiter's-tip posture in a newborn after shoulder dystocia is due to lesion of:

a) C8–T1 b) C5–C6 c) C7 d) T1 only → **Answer: b** (Erb's palsy, upper trunk).

Q2. Foot drop after a fibular-neck injury is due to:

a) Tibial b) Sciatic c) Common peroneal d) Deep peroneal only → **Answer: c**.

Q3. Most medial structure in the femoral canal is:

a) Femoral nerve b) Femoral artery c) Lymph/empty space d) Vein → **Answer: c**.

Q4. Deepest structure in the popliteal fossa:

a) Tibial nerve b) Popliteal vein c) Popliteal artery d) Skin → **Answer: c**.

Q5. Main blood supply to femoral head:

a) Foveal artery b) Medial circumflex femoral c) Femoral artery d) Obturator → **Answer: b**.

Q6. Unlocking of the knee is by:

a) Quadriceps b) Popliteus c) Hamstrings d) Gastrocnemius → **Answer: b**.

Q7. Nerve injured in surgical neck of humerus:

a) Radial b) Axillary c) Musculocutaneous d) Median → **Answer: b**.

Q8. Calcium binds during muscle contraction to:

a) Troponin C b) Tropomyosin c) Actin d) Myosin → **Answer: a**.

Q9. Most important regulator of plasma calcium:

a) Calcitonin b) PTH c) Vit D d) Calcitonin → **Answer: b**.

Q10. 1,25-dihydroxyvitamin D is formed in:

a) Skin b) Liver c) Kidney d) Bone → **Answer: c**.

Q11. Osteoclast formation is stimulated by:

a) OPG b) RANKL c) Calcitonin d) Estrogen → **Answer: b**.

Q12. Saltatory conduction occurs in:

a) Unmyelinated b) Myelinated c) Both d) Axons only → **Answer: b**.

Q13. Articular cartilage is nourished mainly by:

a) Blood vessels b) Synovial fluid diffusion c) Bone marrow d) Lymph → **Answer: b**.

Q14. Osteogenesis imperfecta is a defect of:

a) Type II collagen b) Type I collagen c) Type III d) Type IV → **Answer: b**.

Q15. Hydroxylation of proline in collagen requires:

a) Vit A b) Vit C c) Vit D d) Vit K → **Answer: b**.

Q16. Active vitamin D is:

a) Cholecalciferol b) 25-OH c) 1,25-(OH) $\blacksquare$ d) 24,25-(OH) $\blacksquare$ → **Answer: c**.

Q17. Gout crystals under polarised light are:

a) Positive rhomboid b) Negative needle c) Positive needle d) Neutral → **Answer: b**.

Q18. Hard callus in fracture healing is:

a) Fibrocartilage b) Woven bone c) Lamellar bone d) Fibrous → **Answer: b**.

Q19. Most common organism in acute haematogenous osteomyelitis:

a) Strep b) S. aureus c) E. coli d) Salmonella → **Answer: b**.

Q20. In sickle-cell osteomyelitis, organism commonly:

a) *S. aureus* b) *Salmonella* c) *Strep* d) *Pseudomonas* → **Answer: b.**

Q21. Investigation of choice for early AVN femoral head:

a) X-ray b) MRI c) Bone scan d) CT → **Answer: b.**

Q22. Reye's syndrome is associated with:

a) Paracetamol b) Aspirin in children c) Ibuprofen d) NSAIDs → **Answer: b.**

Q23. Aspirin's antiplatelet effect is due to:

a) Reversible COX-1 b) Irreversible COX-1 c) COX-2 only d) Platelet adhesion → **Answer: b.**

Q24. Atypical femoral fracture is an adverse effect of:

a) Paracetamol b) Bisphosphonates c) Calcitonin d) Vit D → **Answer: b.**

Q25. Most cardiotoxic local anaesthetic:

a) Lidocaine b) Bupivacaine c) Procaine d) Cocaine → **Answer: b.**

Q26. Most common organism in septic arthritis:

a) *N. gonorrhoeae* b) *S. aureus* c) *Strep* d) *E. coli* → **Answer: b.**

Q27. Prophylactic antibiotic for clean ortho implant:

a) Cephalexin oral b) Cefazolin IV c) Amoxicillin d) Metronidazole → **Answer: b.**

Q28. Biofilm protects implant infection by:

a) Killing antibiotics b) Shielding bacteria c) Acid d) Enzymes → **Answer: b.**

Q29. Rigor mortis is due to:

a) ATP excess b) Loss of ATP → actin-myosin lock c) Ca^{2+} loss d) Heat → **Answer: b.**

Q30. Type II collagen is found in:

a) Bone b) Cartilage c) Tendon d) Skin → **Answer: b.**

B. Faculty Surgery (31–55)

Q31. Most common shock in trauma:

a) Cardiogenic b) Hypovolaemic c) Septic d) Neurogenic → **Answer: b.**

Q32. Parkland formula for 50 kg, 40% burn (first 8 h):

a) 2000 b) 4000 c) 8000 d) 1000 mL → **Answer: b** ($4 \times 50 \times 40 = 8000/24h$; half=4000).

Q33. Earliest ECG sign of hyperkalaemia:

a) Wide QRS b) Peaked T c) Sinus brady d) Flat T → **Answer: b.**

Q34. Healing by secondary intention occurs in:

a) Clean sutured wound b) Open granulating wound c) Both d) Neither → **Answer: b.**

Q35. Tetanus toxin acts by:

a) Exciting motor neuron b) Blocking inhibitory GABA/glycine c) Lytic d) Vasospasm → **Answer: b.**

Q36. Passive immunisation in tetanus:

a) TT b) TIG c) Penicillin d) Metronidazole → **Answer: b.**

Q37. Gas gangrene organism:

a) *C. tetani* b) *C. perfringens* c) *Staph* d) *Strep* → **Answer: b.**

Q38. Most common transfusion reaction:

a) Hemolytic b) Febrile non-haemolytic c) TRALI d) Allergic → **Answer: b.**

Q39. TRALI is caused by:

a) Volume overload b) Donor anti-leukocyte antibodies c) ABO d) IgA → **Answer: b.**

Q40. First step in ATLS primary survey:

a) Circulation b) Airway + C-spine c) Exposure d) Disability → **Answer: b.**

Q41. Earliest reliable sign of compartment syndrome:

a) Pulselessness b) Pain on passive stretch c) Pallor d) Paralysis → **Answer: b.**

Q42. Late sign of compartment syndrome:

a) Pain b) Paraesthesia c) Pulselessness/paralysis d) Tense swelling → **Answer: c.**

Q43. Total body surface (both legs, adult, rule of nines):

a) 18 b) 36 c) 27 d) 9 % → **Answer: b.**

Q44. Indication for escharotomy:

a) Superficial burn b) Circumferential full-thickness compromising circulation c) Any burn d) Face → **Answer: b.**

Q45. Suspicious soft-tissue mass should be biopsied by:

a) Excisional b) Planned incisional c) FNA only d) Observe → **Answer: b.**

Q46. Brown tumour of bone is seen in:

a) Hyperparathyroidism b) Osteosarcoma c) GCT d) Metastasis → **Answer: a.**

Q47. Acute limb ischaemia (6 Ps) sudden in AF:

a) Thrombosis b) Embolus c) Vasospasm d) Dissection → **Answer: b.**

Q48. Most common source of arterial embolus:

a) Atheroma b) Atrial fibrillation c) Popliteal aneurysm d) Trauma → **Answer: b.**

Q49. Prophylaxis of DVT after ortho surgery:

a) Aspirin alone b) LMWH + mobilisation + compression c) Warfarin only d) Nothing → **Answer: b.**

Q50. Virchow's triad includes:

a) Stasis, injury, hypercoagulability b) Pain, swelling, redness c) Fever, pus d) None → **Answer: a.**

Q51. Most common level of lower-limb amputation:

a) Above-knee b) Below-knee c) Syme's d) Hip → **Answer: b.**

Q52. Positive Trendelenburg test indicates weakness of:

a) Gluteus maximus b) Gluteus medius c) Quadriceps d) Hamstrings → **Answer: b.**

Q53. Peripheral nerve regeneration rate:

a) 1 cm/day b) 1 mm/day c) 5 mm/day d) 1 mm/month → **Answer: b.**

Q54. Conduction block without axonal disruption:

a) Axonotmesis b) Neurapraxia c) Neurotmesis d) Wallerian → **Answer: b.**

Q55. Kanavel's signs indicate:

a) Felon b) Flexor tenosynovitis c) Paronychia d) Cellulitis → **Answer: b.**

C. Fractures / Trauma (56–80)

Q56. Most common complication of displaced intracapsular neck #:

a) Malunion b) AVN c) Infection d) Stiffness → **Answer: b.**

Q57. Garden Type IV =

a) Incomplete b) Non-displaced complete c) Partial displacement d) Full displacement → **Answer: d.**

- Q58.** Treatment of choice stable intertrochanteric #:
- a) Hemiarthroplasty b) DHS c) Screws d) Cast → **Answer: b.**
- Q59.** Definitive fixation adult femoral shaft #:
- a) Plate b) IM nailing c) Cast d) Traction → **Answer: b.**
- Q60.** Nerve most often injured in supracondylar #:
- a) Radial b) Median (AIN) c) Ulnar d) Musculocutaneous → **Answer: b.**
- Q61.** Dinner-fork deformity =
- a) Smith b) Colles' c) Scaphoid d) Galeazzi → **Answer: b.**
- Q62.** Scaphoid # AVN risk involves:
- a) Waist b) Proximal pole c) Distal pole d) Tuberosity → **Answer: b.**
- Q63.** Schatzker classification is for:
- a) Ankle b) Tibial plateau c) Calcaneus d) Pelvis → **Answer: b.**
- Q64.** Most unstable ankle #:
- a) Weber A b) Weber B c) Weber C d) All equal → **Answer: c.**
- Q65.** Reduced Böhler's angle suggests:
- a) Tibial plateau b) Calcaneal # c) Ankle d) Talar → **Answer: b.**
- Q66.** Most common hip dislocation:
- a) Anterior b) Posterior (90%) c) Central d) Superior → **Answer: b.**
- Q67.** Nerve injured in posterior hip dislocation:
- a) Femoral b) Sciatic c) Obturator d) Pudendal → **Answer: b.**
- Q68.** Unstable vertical-shear pelvic #:
- a) Tile A b) Tile B c) Tile C d) None → **Answer: c.**
- Q69.** Open # with arterial injury:
- a) Gustilo I b) II c) IIIB d) IIIC → **Answer: d.**
- Q70.** Most common Salter-Harris #:
- a) I b) II c) III d) IV → **Answer: b.**
- Q71.** Type V Salter-Harris =
- a) Transverse physis b) Above c) Crush physis d) Through → **Answer: c.**
- Q72.** Fat embolism typically follows:
- a) Skull # b) Long-bone # c) Rib # d) Hand # → **Answer: b.**
- Q73.** Petechiae in fat embolism appear over:
- a) Legs b) Conjunctivae/upper body c) Trunk d) Palms → **Answer: b.**
- Q74.** Highest VTE risk ortho procedure:
- a) Knee arthroscopy b) THR/TKR & pelvis # c) Ankle ORIF d) Hand → **Answer: b.**
- Q75.** Investigation of choice DDH <6 months:
- a) X-ray b) USG c) CT d) MRI → **Answer: b.**
- Q76.** Treatment of choice idiopathic clubfoot:
- a) Surgery b) Ponseti casting c) Splint d) Observe → **Answer: b.**
- Q77.** SCFE is shown by:
- a) Klein's line not crossing head b) Trethowan c) Shenton d) Skinner → **Answer: a.**

Q78. Treatment of stable SCFE:

a) Manipulative reduction b) In-situ pinning c) Cast d) Observe → **Answer: b.**

Q79. Rachitic rosary is seen in:

a) Osteomalacia b) Rickets c) Scurvy d) OI → **Answer: b.**

Q80. Looser's zones are characteristic of:

a) Rickets b) Osteomalacia c) OA d) Osteoporosis → **Answer: b.**

D. Paediatric Orthopaedics (81–100)

Q81. Ortolani sign means:

a) Dislocatable b) Reducible click c) Painful d) Fixed → **Answer: b.**

Q82. Barlow test detects:

a) Reducible b) Dislocatable hip c) Fixed d) Normal → **Answer: b.**

Q83. Most common CP type:

a) Ataxic b) Dyskinetic c) Spastic d) Mixed → **Answer: c.**

Q84. Botulinum in CP treats:

a) Spasticity b) Weakness c) Pain d) Seizure → **Answer: a.**

Q85. Greatest lower-limb growth from:

a) Proximal femur b) Distal femur + proximal tibia c) Tibial shaft d) Fibula → **Answer: b.**

Q86. Perthes occurs typically in:

a) <2 yr b) Boys 4–8 yr c) Adolescents d) Adults → **Answer: b.**

Q87. Lateral-pillar (Herring) C implies:

a) Excellent b) Poor prognosis c) Mild d) Stable → **Answer: b.**

Q88. Blue sclerae characteristic of:

a) Rickets b) OI c) Marfan d) Achondroplasia → **Answer: b.**

Q89. Achondroplasia causes:

a) Membranous defect b) Endochondral defect (FGFR3) c) Vit D d) Trauma → **Answer: b.**

Q90. In achondroplasia, limbs are:

a) Long b) Short (rhizomelic) c) Normal d) Bowed only → **Answer: b.**

Q91. Congenital pseudoarthrosis most common in:

a) Femur b) Tibia c) Humerus d) Radius → **Answer: b.**

Q92. Associated condition with congenital pseudoarthrosis tibia:

a) OI b) NF1 c) Rickets d) CPA → **Answer: b.**

Q93. Most common bone tumour in children:

a) Ewing's b) Osteosarcoma c) Metastasis d) GCT → **Answer: b.**

Q94. Most common malignant bone tumour overall:

a) Osteosarcoma b) Metastasis c) Ewing's d) Myeloma → **Answer: b.**

Q95. Most common fracture in osteoporosis:

a) Hip b) Vertebral > Colles' > hip c) Rib d) Skull → **Answer: b.**

Q96. Most common cause of AVN femoral head:

a) Alcohol b) Steroids/trauma c) Infection d) Age → **Answer: b.**

Q97. Most common site paediatric haematogenous osteomyelitis:

a) Diaphysis b) Metaphysis (distal femur) c) Epiphysis d) Apophysis → **Answer: b.**

Q98. Sequestrum is:

a) New bone shell b) Dead bone c) Pus d) Granulation → **Answer: b.**

Q99. Involucrum is:

a) Dead bone b) New living shell c) Pus d) Sinus → **Answer: b.**

Q100. Most common congenital ortho anomaly:

a) DDH b) CTEV (clubfoot) c) Polydactyly d) Syndactyly → **Answer: b** (CTEV most common overall; DDH most common hip).

E. Spine (101–115)

Q101. Ipsilateral weakness + contralateral pain/temp loss:

a) Central cord b) Brown-Séquard c) Anterior cord d) Cauda → **Answer: b.**

Q102. Hyperextension in cervical spondylosis →

a) Brown-Séquard b) Central cord c) Anterior cord d) Cauda → **Answer: b.**

Q103. Most common level of lumbar disc prolapse:

a) L3–L4 b) L4–L5 c) L5–S1 d) L2–L3 → **Answer: b.**

Q104. Positive straight-leg raise indicates:

a) Spinal stenosis b) Lumbar disc prolapse c) Spondylolisthesis d) OA → **Answer: b.**

Q105. Isthmic spondylolisthesis is due to:

a) Degeneration b) Pars interarticularis defect c) Trauma d) Infection → **Answer: b.**

Q106. Meyerding grade III slip =

a) ≤25% b) 25–50% c) 50–75% d) >75% → **Answer: c.**

Q107. Scoliosis is measured by:

a) Cobb angle b) Schober c) Bohler d) Klein → **Answer: a.**

Q108. Bracing indicated in AIS:

a) <10° b) 25–45° with growth c) >60° d) Any → **Answer: b.**

Q109. Scheuermann's kyphosis shows:

a) Wedging of ≥3 vertebrae b) Disc bulge c) Listhesis d) Sclerosis → **Answer: a.**

Q110. Angular kyphosis with cold abscess suggests:

a) Osteoporosis b) Pott's (TB) c) Scheuermann d) OA → **Answer: b.**

Q111. Spinal TB spreads by:

a) Haematogenous to body then paradiscal b) Direct cord c) Lymph d) Skin → **Answer: a.**

Q112. Neurogenic claudication relieved by:

a) Rest b) Flexion/forward bend c) Standing d) Walking → **Answer: b.**

Q113. Cauda equina includes:

a) Back pain only b) Saddle anaesthesia + urinary retention c) Fever d) Sciatica → **Answer: b.**

Q114. Management of cauda equina:

a) Physio b) Urgent decompression c) NSAIDs d) Observe → **Answer: b.**

Q115. Schober test <4 cm indicates:

a) Normal b) Limited lumbar flexion (AS) c) Scoliosis d) Listhesis → **Answer: b.**

F. Joints / Arthritis (116–135)

Q116. Radiologic feature of OA:

a) JSN + osteophytes b) Erosions c) Soft-tissue d) Periostitis → **Answer: a.**

Q117. Kellgren-Lawrence classifies:

a) RA b) OA severity c) Spondylolisthesis d) Scoliosis → **Answer: b.**

Q118. Cervical subluxation is complication of:

a) OA b) RA c) AS d) Gout → **Answer: b.**

Q119. Swan-neck deformity =

a) PIP flex + DIP ext b) PIP ext + DIP flex c) Both flex d) Both ext → **Answer: b.**

Q120. RA spares:

a) MCP b) PIP c) DIP d) Wrist → **Answer: c.**

Q121. Investigation of choice septic arthritis:

a) X-ray b) Aspiration + culture c) ESR d) USG → **Answer: b.**

Q122. Most common organism septic arthritis:

a) Gonococcus b) S. aureus c) Strep d) E. coli → **Answer: b.**

Q123. Gout crystals are:

a) Positive rhomboid b) Negative needle c) Positive needle d) Neutral → **Answer: b.**

Q124. Pseudogout crystals are:

a) Calcium pyrophosphate (positive) b) Urate c) CPP negative d) Cholesterol → **Answer: a.**

Q125. Bamboo spine characteristic of:

a) RA b) AS c) OA d) PsA → **Answer: b.**

Q126. HLA-B27 associated with:

a) RA b) AS c) OA d) Gout → **Answer: b.**

Q127. Gold-standard early AVN:

a) X-ray b) MRI c) Bone scan d) CT → **Answer: b.**

Q128. Charcot joint due to:

a) Infection b) Neuropathy c) Trauma d) Crystal → **Answer: b.**

Q129. Most common cause Charcot today:

a) Syphilis b) Diabetes c) SCI d) Leprosy → **Answer: b.**

Q130. Hemophilic arthropathy from:

a) Inflammation b) Recurrent haemarthrosis c) Infection d) Crystal → **Answer: b.**

Q131. Target joints in haemophilia:

a) Shoulder/knee b) Knee/elbow/ankle c) Hip/spine d) Hand → **Answer: b.**

Q132. Most common joint replaced:

a) Hip b) Knee > hip c) Shoulder d) Ankle → **Answer: b.**

Q133. Osteoporosis DEXA T-score ≤:

a) -1 b) -2.5 c) -3.5 d) 0 → **Answer: b.**

Q134. Osteopenia T-score range:

a) -2.5 to -3.5 b) -1 to -2.5 c) >-1 d) <-2.5 → **Answer: b.**

Q135. First-line osteoporosis:

a) Calcitonin b) Bisphosphonate c) Strontium d) Vit D only → **Answer: b.**

G. Sports / Soft Tissue (136–145)

Q136. Most sensitive test for ACL tear:

a) Anterior drawer b) Lachman c) Pivot shift d) McMurray → **Answer: b.**

Q137. Mechanism ACL injury:

a) Hyperflexion b) Non-contact valgus rotation c) Direct blow d) Fall → **Answer: b.**

Q138. Bucket-handle meniscal tear causes:

a) Click b) Locked knee c) Swelling d) Bruise → **Answer: b.**

Q139. McMurray test assesses:

a) ACL b) Meniscus c) PCL d) Patella → **Answer: b.**

Q140. Most commonly torn rotator cuff tendon:

a) Subscapularis b) Supraspinatus c) Infraspinatus d) Teres → **Answer: b.**

Q141. Painful arc $60-120^\circ$ suggests:

a) Frozen shoulder b) Supraspinatus impingement c) AC arthritis d) OA → **Answer: b.**

Q142. Tennis elbow involves:

a) Medial epicondyle b) Lateral epicondyle (ECRB) c) Olecranon d) Radial head → **Answer: b.**

Q143. Positive Thompson test (no plantarflex on squeeze) =

a) Calf strain b) Achilles rupture c) DVT d) Compartment → **Answer: b.**

Q144. Drop-arm test indicates:

a) AC joint b) Supraspinatus/large cuff tear c) Frozen d) OA → **Answer: b.**

Q145. Most commonly injured meniscus:

a) Lateral b) Medial c) Both equal d) Discoid → **Answer: b.**

H. Tumours / Infection (146–165)

Q146. Sunburst appearance on X-ray =

a) Ewing's b) Osteosarcoma c) GCT d) Metastasis → **Answer: b.**

Q147. Codman triangle seen in:

a) Ewing's b) Osteosarcoma c) GCT d) Osteoid osteoma → **Answer: b.**

Q148. Osteosarcoma arises in:

a) Diaphysis b) Metaphysis c) Epiphysis d) Flat bone → **Answer: b.**

Q149. Onion-skin periosteal reaction =

a) Osteosarcoma b) Ewing's c) GCT d) Chondrosarcoma → **Answer: b.**

Q150. Ewing's translocation:

a) $t(11;22)$ b) $t(9;22)$ c) $t(8;14)$ d) $t(15;17)$ → **Answer: a.**

Q151. Ewing's arises in:

a) Metaphysis b) Diaphysis c) Epiphysis d) Spine only → **Answer: b.**

Q152. Sclerotic metastasis suggests:

a) Breast b) Prostate c) Lung d) Kidney → **Answer: b.**

Q153. Lytic metastasis common from:

a) Prostate b) Breast/lung c) Thyroid d) All blastic → **Answer: b.**

Q154. Soap-bubble appearance =

a) Osteosarcoma b) GCT c) Ewing's d) ABC → **Answer: b.**

Q155. GCT occurs in:

a) Metaphysis b) Epiphysis (around knee) c) Diaphysis d) Skull → **Answer: b.**

Q156. Campanacci grading is for:

a) Osteosarcoma b) GCT c) Ewing's d) Chondrosarcoma → **Answer: b.**

Q157. Soft-tissue sarcoma biopsy principle:

a) Excisional first b) Planned incisional then wide excision c) FNA only d) Observe → **Answer: b.**

Q158. Most common site of sarcoma metastasis:

a) Liver b) Lung c) Bone d) Brain → **Answer: b.**

Q159. Acute osteomyelitis in kids localises to:

a) Diaphysis b) Metaphysis c) Epiphysis d) Apophysis → **Answer: b.**

Q160. Most common organism in osteomyelitis:

a) Strep b) *S. aureus* c) *Salmonella* d) *Pseudomonas* → **Answer: b.**

Q161. In sickle-cell osteomyelitis organism:

a) *S. aureus* b) *Salmonella* c) *Staph epidermidis* d) Anaerobe → **Answer: b.**

Q162. Brodie's abscess is:

a) Acute b) Subacute localised osteomyelitis c) Chronic d) Tumour → **Answer: b.**

Q163. Most common organism septic arthritis neonate:

a) *S. aureus* b) GBS/*E. coli* c) *Gonococcus* d) Strep → **Answer: b.**

Q164. Pott's disease organism:

a) Staph b) *M. tuberculosis* c) *Salmonella* d) *E. coli* → **Answer: b.**

Q165. Implant infection protected by:

a) Pus b) Biofilm c) Capsule d) Cement → **Answer: b.**

I. Image-Based / Investigations (166–180)

Q166. Investigation of choice for AVN:

a) X-ray b) MRI c) Bone scan d) CT → **Answer: b.**

Q167. Investigation of choice for DDH <6 mo:

a) X-ray b) USG c) MRI d) CT → **Answer: b.**

Q168. Best modality for soft tissue/meniscus:

a) X-ray b) MRI c) Bone scan d) USG → **Answer: b.**

Q169. Best for complex pelvic/acetabular #:

a) X-ray b) CT c) MRI d) Bone scan → **Answer: b.**

Q170. Bone scan is best for:

a) Acute # b) Occult #/metastases/stress # c) Meniscus d) Ligament → **Answer: b.**

Q171. X-ray of Colle's shows:

a) Volar angulation b) Dorsal angulation (dinner fork) c) Normal d) Shortening only → **Answer: b.**

Q172. X-ray of NOF # with displaced femoral head suggests:

a) Intertrochanteric b) Intracapsular Garden IV c) Subtrochanteric d) Stress → **Answer: b.**

Q173. Scoliosis screening measurement:

a) Schober b) Cobb c) Bohler d) Klein → **Answer: b.**

Q174. Calcaneal # X-ray sign:

a) Increased Bohler b) Decreased Böhler ($<20^\circ$) c) Normal d) Step → **Answer: b.**

Q175. Supracondylar # X-ray shows:

a) Dorsal displacement b) Posterior displacement of distal fragment c) Normal d) Oblique → **Answer: b.**

Q176. Osteosarcoma X-ray classic:

a) Lytic only b) Sunburst + Codman c) Soap-bubble d) Onion-skin → **Answer: b.**

Q177. Ewing's X-ray classic:

a) Sunburst b) Onion-skin c) Soap-bubble d) Blastic → **Answer: b.**

Q178. GCT X-ray classic:

a) Sunburst b) Soap-bubble (epiphysis) c) Onion-skin d) Blastic → **Answer: b.**

Q179. Pott's spine X-ray shows:

a) Listhesis b) Vertebral collapse + paraspinal abscess c) Normal d) Scoliosis only → **Answer: b.**

Q180. Most common bone tumour in children overall:

a) Ewing's b) Osteosarcoma c) Metastasis d) Lymphoma → **Answer: b.**

Next: `09-RAPID-REVISION-EXAM-NIGHT.md` (rapid revision, flowcharts, image-based, exam-night chart, 3-month plan).

PART 5 — RAPID REVISION · FLOWCHARTS · IMAGE-BASED · EXAM-NIGHT CHART · 3-MONTH PLAN

5A. RAPID REVISION (one-liners)

Nerves: Erb C5–C6 (waiter's tip) · Klumpke C8–T1 (claw+Horner) · Axillary = surgical neck → deltoid · Radial = spiral → wrist drop · Median (AIN) = supracondylar → no OK sign · Common peroneal = fibular neck → foot drop · Sciatic = post hip dislocation.

Hip blood: MCFA retinacular = head supply; intracapsular # → AVN.

Calcium: PTH↑Ca, calcitonin↓Ca, vit D = gut absorption; active = 1,25-OH (kidney).

Bone tumours: Osteo=metaphysis/sunburst/Codman · Ewing=diaphysis/onion/t(11;22) · GCT=epiphysis/soap-bubble · Metastasis=most common (prostate blastic, breast/lung lytic).

Classifications: Garden (NOF) · Gustilo (open) · Salter-Harris (physeal) · Weber (ankle) · Gartland (supracondylar) · Schatzker (plateau) · Evans (intertroch) · Cobb (scoliosis) · Meyerding (spondylolisthesis) · Tile (pelvis) · Herring (Perthes) · Campanacci (GCT) · Klein (SCFE) · Böhler (calcaneus).

"Most common": organism = *S. aureus* · child bone tumour = osteosarcoma · adult bone tumour = metastasis · osteo site kids = metaphysis · AVN = NOF/trauma/steroids · congenital = CTEV · limp child = transient synovitis · joint replaced = knee>hip.

Red flags: Cauda equina (saddle + retention) · compartment (pain+passive stretch; pulseless=LATE) · gas gangrene (crepitus) · Pott's (cold abscess+angular kyphosis) · septic arthritis (hot joint → aspirate first).

5B. FLOWCHARTS (ASCII)

Acute swollen joint:

Swollen joint ■ Hot/red/fever? ■ YES → Septic arthritis → ASPIRATE before ABx → drain+ABx
 ■ NO
 ■ Monoarticular? ■ Gout (neg needle, 1st MTP) / Pseudogout (pos, knee)
 Polyarticular? ■ RA (MCP/PIP, swan-neck) / AS (bamboo, HLA-B27)

Child with limp:

Limp ■ <3 mo: Barlow/Ortolani? ■ YES → DDH
 ■ NO
 ■ Fever+localised thigh pain → Osteomyelitis / Septic arthritis (aspirate)
 ■ 4-8 yr, painless limp → Perthes (AVN)
 ■ Obese adolescent, knee→hip → SCFE (Klein)
 ■ No fever, self-limited → Transient synovitis (most common)
 ■ Foot deformity → CTEV

Suspected compartment syndrome:

Pain + tense swelling + pain on PASSIVE STRETCH → measure pressure (>30 mmHg) → URGENT FASCIOTOMY
 (Do NOT wait for pulselessness/paralysis – those are LATE)

Open fracture:

Open # ■ Tetanus status ■ ABx (cefazolin±amino/pen) within 1 h ■ Debridement <6-8 h
 ■ Stabilise (fixator/ORIF) ■ Soft-tissue cover

Hip pain in elderly after fall:

X-ray ■ NOF (intracapsular) displaced → AVN risk → arthroplasty
 ■ Intertrochanteric → DHS/IM nail (good union)
 ■ Posterior dislocation → reduce <6-12 h (sciatic nerve)

5C. IMAGE-BASED REVISION (X-ray / scan "spot" list)

Image	Key sign to spot
Colles'	Dorsal angulation → dinner fork
Smith	Volar angulation (reverse)
NOF Garden IV	Displaced femoral head, fallen fragment
Supracondylar	Posterior displacement, elbow
Scaphoid	Snuffbox tenderness; X-ray may miss → MRI
Scoliosis	Cobb angle >10°
TB spine (Pott's)	Vertebral collapse + paraspinal cold abscess
Osteosarcoma	Sunburst + Codman triangle, metaphysis
Ewing's	Onion-skin, diaphysis
GCT	Soap-bubble, epiphysis around knee
Calcaneus	Böhler's angle <20°
Knee OA	Joint-space narrowing + osteophytes
RA hand	Erosions, ulnar deviation, swan-neck
AS spine	Bamboo spine, syndesmophytes

5D. EXAM-NIGHT FOCUS CHART (must revise tonight)

Priority	Topic cluster	Action
★★★★★	Classifications (Garden, Gustilo, Salter-Harris, Weber, Gartland, Schatzker, Cobb, Evans, Tile, Klein, Böhler)	Flash-cards
★★★★★	Nerve injuries (radial/ axillary/ median-AIN/ common peroneal/ sciatic)	Match lesion→deficit
★★★★★	Paediatric hip (DDH, Perthes, SCFE, transient synovitis)	Ortolani/Barlow, Klein, Herring
★★★★★	Bone tumours (osteoid/Ewing/GCT/metastasis signs)	Imaging sign→Dx
★★★★★	Osteomyelitis/septic arthritis (S. aureus, aspirate first)	Organism + step
★★★★★	Compartment syndrome & cauda equina (RED FLAGS)	Don't wait for late signs
★★★★★	Arthritis (OA/RA/AS/gout/pseudogout)	X-ray/crystal/HLA
★★★★★	Spine (disc/SLR, TB, stenosis, scoliosis)	SLR, Cobb, Schober
★★★★★	AVN (MRI, causes) · Osteoporosis (DEXA) · Fracture healing	Investigation-of-choice
★★★	Faculty surgery (shock, tetanus, DVT, burns, trauma)	Numbers (Parkland, 5/6 Ps)
★★★	Basic sciences (calcium, collagen, vitamin D, NSAIDs)	Mechanism stems

5E. 3-MONTH STUDY PLAN (for a sure chance)

Daily routine (every day): 20 SBA drill (Part 4) · 1 X-ray/image (5C) · 30 min revision of ★★★★★ topics.

Week	Focus	Milestone
1–2	Basic Sciences (Anatomy/Physio/Biochem/Path/Pharma/Micro) — topics #1–25	Finish + self-test 50 Q
3–4	Faculty Surgery #26–45 + trauma/ATLS	OSCE-style nerve/vessel recall
5–6	Ortho fractures/trauma #46–60 + paediatric #61–70	Classification flash-cards done
7	Spine #71–78 + joints/arthritis #79–86	SLR/Schober/Cobb drill
8	Sports/soft tissue #87–90 + tumours/infection #91–96	Imaging sign→Dx deck
9	Principles #97–100 + Part 4 full bank (round 1)	180-Q mock (timed)
10	Weak-area loop + repeat bank (round 2)	Mock 2
11	Rapid revision (5A/5B/5C) + exam-night chart	Mock 3 + error log
12	Taper: only ★★★★★ + image list + sleep	Exam day

Weekly weightage: Ortho 40% · Basic Sciences 33% · Surgery 27%.

Mock tests: Weeks 9, 10, 11 (100 Q, 1 h each) → target ≥75% before exam.

Error log: keep a notebook of every wrong answer → revise those topics first next day.

Final step: Assemble `MASTER-DIPLOMA-ORTHOPAEDICS-2027.md` + compile print-ready PDF via `build/build\_pdf.py`.

APPENDIX A — GEMINI DEEP-RESEARCH COMMAND

GEMINI DEEP RESEARCH COMMAND (Professional / Copy-Paste Ready)

Use this prompt in Google Gemini (gemini.google.com → "Deep Research") to independently validate & enrich this book

INSTRUCTIONS TO USER: Copy everything between the two `=====` lines and paste into Gemini Deep Research. Replace bracketed `[...]` fields with your specific exam session if needed.

=====
ROLE

You are a senior medical education researcher and orthopaedic residency examiner with 20+ years of experience preparing candidates for postgraduate entrance examinations in Bangladesh.

OBJECTIVE

Produce a rigorous, evidence-based research dossier that validates and extends a "Master Diploma Orthopaedics Admission Preparation Guide (Bangladesh, 2027 Edition)" for the Diploma in Orthopaedics entrance examination conducted by Bangladesh Medical University (BMU) / Bangabandhu Sheikh Mujib Medical University (BSMMU), including affiliated institutes and NITOR.

CONTEXT

- Target exam: Diploma in Orthopaedics (D.Ortho) admission MCQ test, academic session July 2026 / 2027 intake.
- Parallel exams to benchmark: BSMMU MD/MS (Residency) Orthopaedics written MCQ (session up to March 2026) and BMU M.Phil Orthopaedics.
- No single fixed public syllabus is published; the test draws from MBBS Basic Sciences + Faculty of Surgery + Orthopaedics.
- Candidate profile: MBBS graduates; need high-yield, probability-ranked topics, previous-question trends, SBA/MCQ patterns, and references.

RESEARCH TASKS (perform all, cite sources)

- T1. Recover and summarise the official BMU/BSMMU Diploma & M.Phil admission notices for July 2026: eligibility, seat matrix, exam format (marks, number of MCQs, negative marking), and timeline.
- T2. Recover BSMMU Residency (Orthopaedics) admission MCQ pattern up to March 2026: subject weightage, question style, recurring themes.
- T3. Mine publicly available previous-year MCQ/SBA from BMU, BSMMU, NITOR, and Bangladesh PG entrance banks; extract the TOP 100 most frequently repeated orthopaedic + basic-science + surgery topics.
- T4. For each of those 100 topics, assign a yield probability (Very High / High / Moderate / Low / Rare) with the rationale based on repetition frequency and syllabus weighting.
- T5. Identify examiner-favourite SBA stems (e.g., "All are true EXCEPT", "Most common...", "Best investigation...", "First-line management...") and the classic distractors used in Bangladesh.
- T6. Cross-check every factual claim against standard references: Apley, Campbell, Rockwood & Green, Miller's Review, AO Principles (Ortho); Bailey & Love, Schwartz, Sabiston (Surgery); Gray's, Snell, Ganong, Guyton, Robbins, Katzung, Lippincott (Basic Sciences).
- T7. Build a "Previous-Question Trend Analysis" table: topic → years seen → variant asked → predicted 2027 probability.
- T8. Flag any Bangladesh-specific epidemiology (e.g., road-traffic trauma, open fractures, spinal TB / Pott's, vitamin-D deficiency, rickets, osteomyelitis patterns) that raises yield.

SOURCES (prioritise, in order)

1. bmu.ac.bd and bsmmu.edu.bd official notices
2. NITOR (nitor.gov.bd) academic resources
3. Bangladesh PG medical entrance question banks (doctorsgang, medicos community repos, Bengali medical coaching sites)
4. Standard textbooks listed above (verify facts)
5. Cochrane / PubMed for management controversies
6. Orthopaedic society guidelines (AAOS, BOA, BOS)

CONSTRAINTS

- Cite every statistic or claim with [source name / URL].
- Prefer 2020–2026 evidence; note if a fact is older but still standard.
- Distinguish "verified from official notice" vs "derived from question banks / examiner lore" – do NOT overstate certainty.
- Output in Markdown, Bangladesh/Bengali-friendly phrasing allowed.
- Flag anything that contradicts this book so the author can correct it.

OUTPUT FORMAT

1. Executive summary (bullet points).
2. Exam-pattern table (BMU Diploma July 2026 vs BSMMU Residency Mar 2026).
3. Subject weightage estimate (Basic Sciences / Surgery / Ortho %).
4. Top-100 topic table: # | Topic | Category | Repeat count | Probability | Key refs.
5. Examiner-favourite SBA stem library (≥30 stems with model answers).
6. Previous-question trend analysis table.
7. Gaps / contradictions vs the existing draft book.
8. Recommended 3-month study weightage per subject.

VALIDATION CRITERIA

- Every probability rating has a stated reason.
- No un-cited "100% common" claim without evidence.
- At least 15 distinct web/PDF sources referenced.
- Output is directly usable to update the preparation book.

=====

How to use this with the rest of the book

- Paste into Gemini → let it return the dossier → cross-check against `04-TOP-100-INDEX.md`.
- Where Gemini's probability differs from ours, trust the **higher-confidence** source and note it in your revision margin.
- Use Gemini's "Examiner-favourite SBA stem library" to expand `08-SBA-MCQ-BANK.md`.